

POLITECNICO
MILANO 1863

DEPARTMENT OF MECHANICAL
ENGINEERING

EXERCISE CLASS 5

Linear models

Name Surname



DIPARTIMENTO DI ECCELLENZA
MIUR 2018-2022



Milano

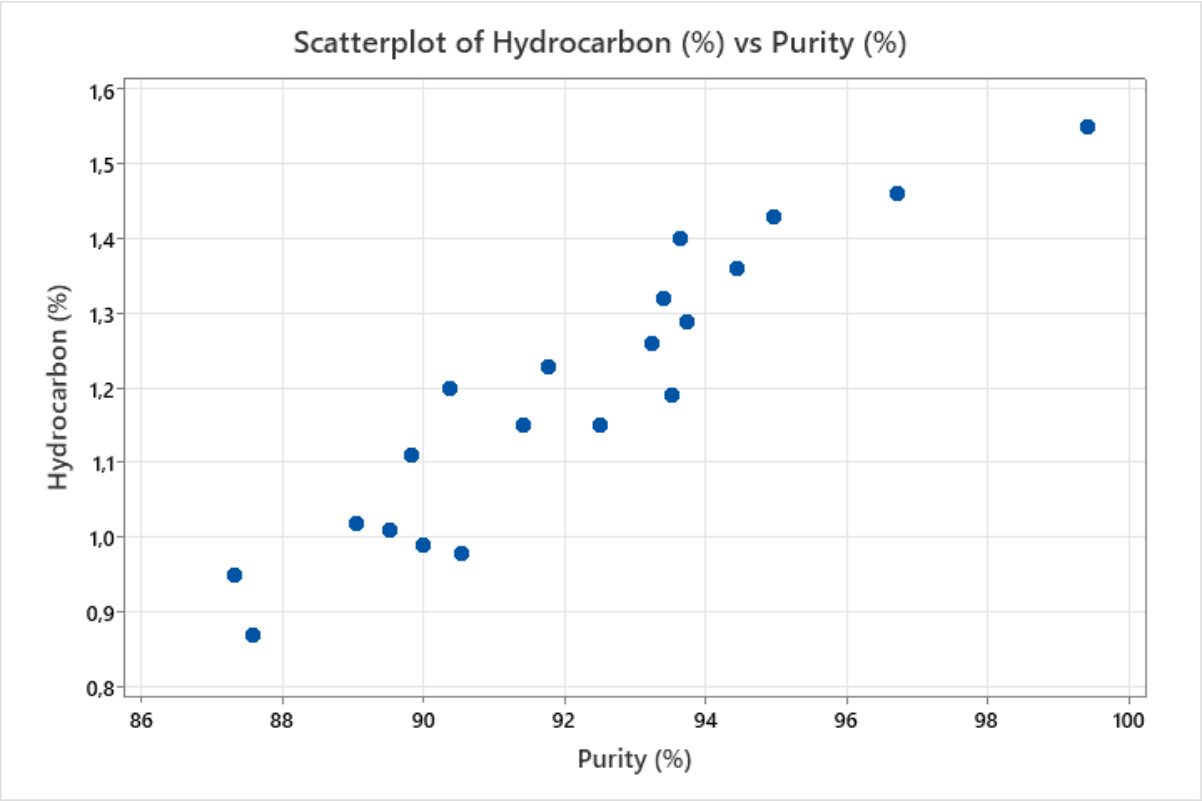
Exercise 1

Consider the data in `oxygen\_hydrocarbon\_levels.csv`. In this data frame, y is the purity of oxygen produced in a chemical distillation process, and x is the percentage of hydrocarbons present in the main condenser of the distillation unit. The data were sampled automatically every hour during plant operation and are in chronological order.

1. Fit a simple linear regression model to the oxygen purity.
2. Compute the 95% confidence interval on the slope parameter.
3. Suppose that we are interested in predicting mean oxygen purity when the hydrocarbon level is 1.00%. Compute the 95% confidence interval on the mean response.
4. Suppose that we are interested in predicting mean oxygen purity when the hydrocarbon level is 1.00%. Compute the 95% prediction interval on the process outcome.

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

It looks like there is a strong correlation between the two variables. Let's plot the purity of oxygen (y), against the percentage of hydrocarbons (x): [Graph](#) → [Scatterplot](#) → [Simple](#)



There is a clear **linear correlation** between the two variables.
Therefore, Hydrocarbon (%) is a promising **regressor** to predict Purity (%)

Short reminder about regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim iid(\mathbf{0}, \sigma^2 \mathbf{I})^*$$

$\mathbf{y}$ is $n \times 1$

$\mathbf{X}$ is $n \times p$ (p = number of parameters), e.g.: $y = \beta_0 + \beta_1 x + \varepsilon \rightarrow p=2$

$\boldsymbol{\beta}$ is $p \times 1$

$\boldsymbol{\varepsilon}$ is $n \times 1$

$$\mathbf{X} = \begin{bmatrix} \mathbf{1} & x_{11} & \dots & x_{1k} \\ \dots & \dots & \dots & \dots \\ \mathbf{1} & x_{n1} & \dots & x_{nk} \end{bmatrix} \quad p = k + 1$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Estimated parameters

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$$

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

Estimated response

$$V(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

$$\hat{\sigma}^2 = MS_E$$

Estimated variance

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$$

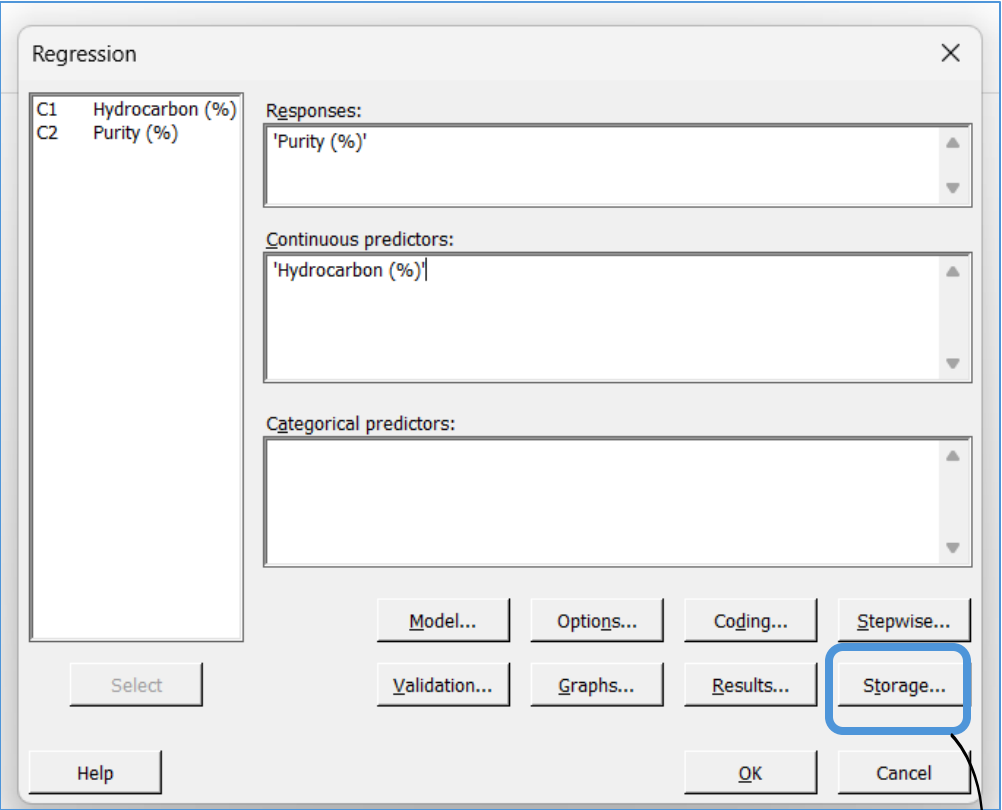
Residuals

* But to make inference we need normality assumption (NID)

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

Let’s calculate a regression model using **hydrocarbon** as the independent variable and **purity** as the dependent variable:

Stat → Regression → Regression → Fit regression model



Tick “Fits” and “Residuals”

Regression Equation

Purity (%) = 74,28 + 14,95 Hydrocarbon (%)

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	74,28	1,59	46,62	0,000	
Hydrocarbon (%)	14,95	1,32	11,35	0,000	1,00

Model Summary

σ_ϵ	S	R-sq	R-sq(adj)	R-sq(pred)
	1,08653	87,74%	87,06%	84,13%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	152.127	152.127	128.86	0.000
Hydrocarbon (%)	1	152.127	152.127	128.86	0.000
Error	18	21.250	1.181		
Lack-of-Fit	17	20.656	1.215	2.05	0.506
Pure Error	1	0.594	0.594		
Total	19	173.377			

Both parameters are statistically significant. Now, we must check the normality and randomness assumption on residuals.

Short reminder about regression

Test for significance of regression

$$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1: \exists \beta_i \neq 0$$

(usually we exclude the constant term)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	152.127	152.127	128.86	0.000
Hydrocarbon (%)	1	152.127	152.127	128.86	0.000
Error	18	21.250	1.181		
Lack-of-Fit	17	20.656	1.215	2.05	0.506
Pure Error	1	0.594	0.594		
Total	19	173.377			

Partition of the variance: $SS_{TOT} = SS_{REG} + SS_E$

Degrees of freedom: $n - 1$ $p - 1$ $n - p$

E.g., simple regression:

$$SS_{TOT} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SS_{REG} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SS_E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Short reminder about regression

Test for significance of regression

$$H_0: \beta_1 = \beta_1 = \dots = \beta_k = 0$$

$$H_1: \exists \beta_i \neq 0$$

(usually we exclude the constant term)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
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Hydrocarbon (%)	1	152.127	152.127	128.86	0.000
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Lack-of-Fit	17	20.656	1.215	2.05	0.506
Pure Error	1	0.594	0.594		
Total	19	173.377			

Partition of the variance: $SS_{TOT} = SS_{REG} + SS_E$

Degrees of freedom: $n - 1$ $p - 1$ $n - p$

Reminder:

$$\frac{(n - 1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

If H_0 is true: $\frac{SS_{REG}}{\sigma^2} \sim \chi_{p-1}^2$ and $\frac{SS_E}{\sigma^2} \sim \chi_{n-p}^2$ then:

$$F_0 = \frac{SS_{REG}/(p - 1)}{SS_E/(n - p)} = \frac{MS_{REG}}{MS_E} \sim F(p - 1, n - p)$$

Reject H_0 if: $F_0 > F_\alpha(p - 1, n - p)$

Short reminder about regression

Test for individual coefficients

$$H_0: \beta_i = 0$$

$$H_1: \beta_i \neq 0$$

Test statistic: $T_0 = \frac{\hat{\beta}_i}{se(\hat{\beta}_i)} = \frac{\hat{\beta}_i}{\sqrt{\hat{\sigma}^2 C_{ii}}}$ where C_{ii} is the i -th element on the diagonal of $(\mathbf{X}^T \mathbf{X})^{-1}$

If H_0 is true: $T_0 \sim t_{n-p}$

Reject H_0 if: $|T_0| > t_{\frac{\alpha}{2}, n-p}$

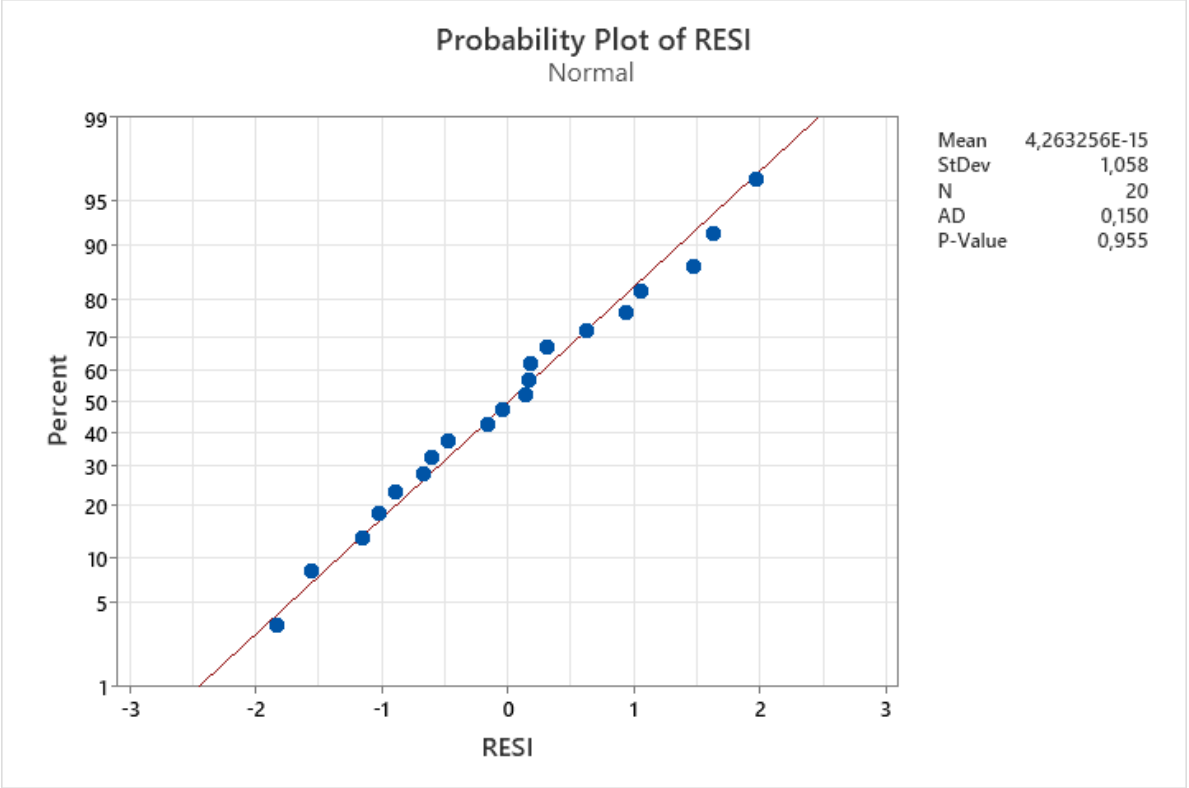
Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	74,28	1,59	46,62	0,000	
Hydrocarbon (%)	14,95	1,32	11,35	0,000	1,00

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

Assumption on residuals: Normality

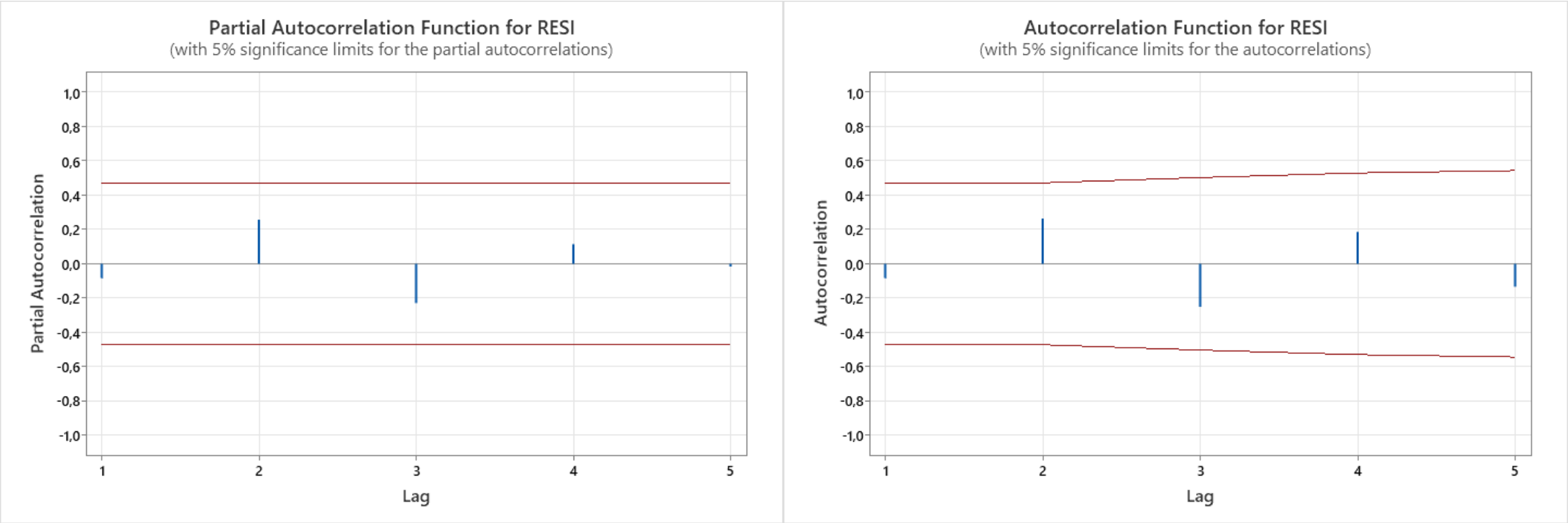
Stat → Basic statistics → Normality test



We fail to reject the null hypothesis of normality, so we can assume that the residuals are normally distributed.

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

Assumption on residuals: randomness (to be checked because data are reported chronologically)



Runs test:

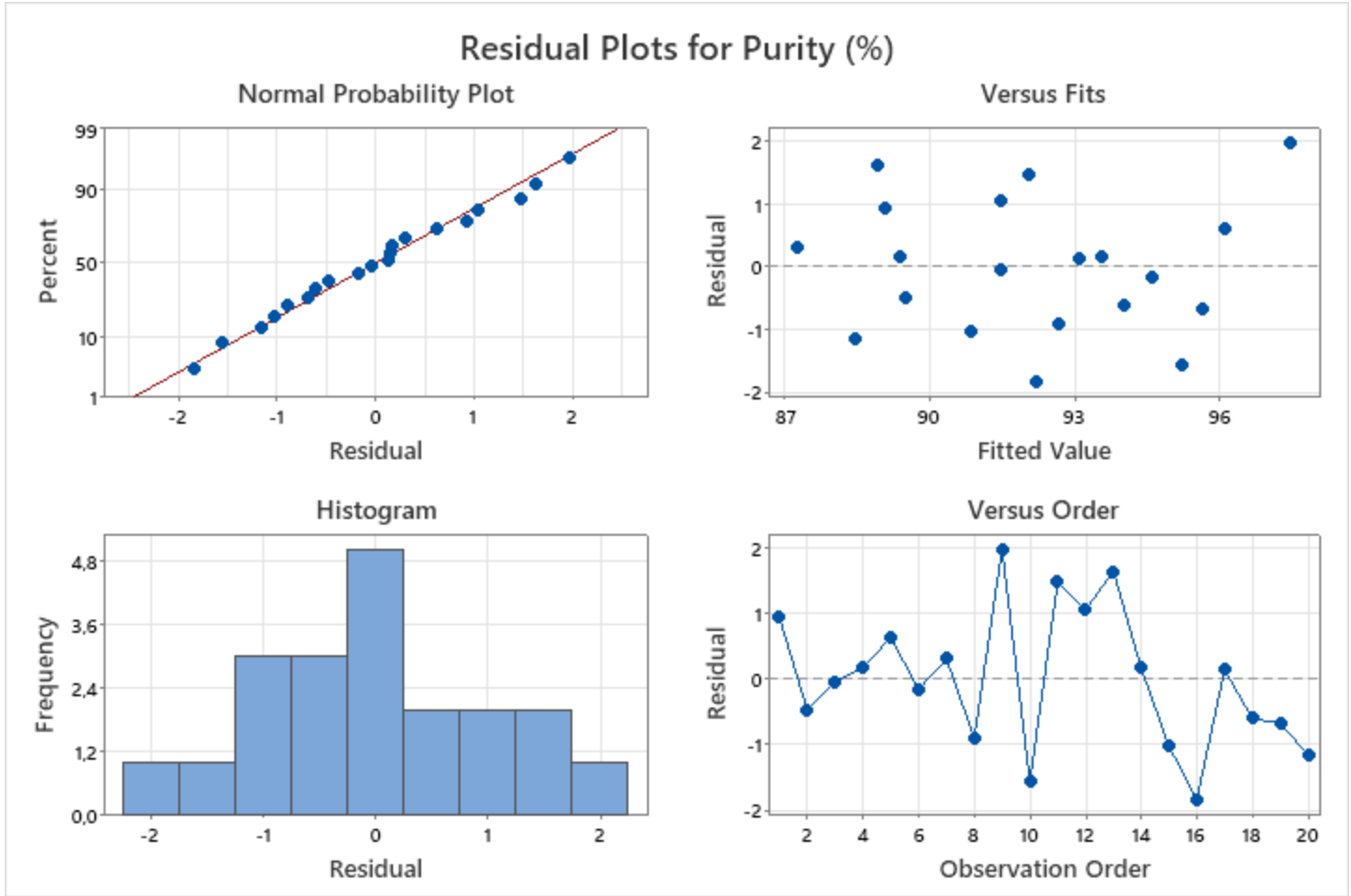
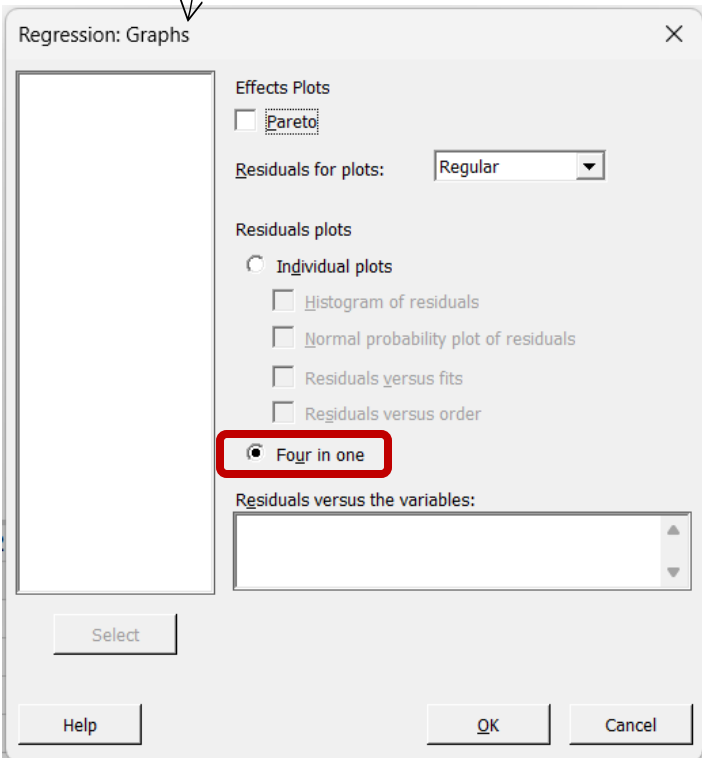
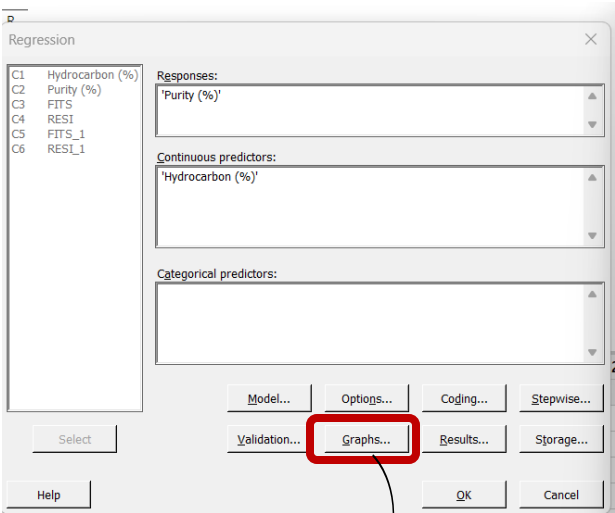
Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
12	11,00	0,646

Residuals do not show autocorrelation and we accept the null hypothesis of the runs-test.

Assumption on randomness of residuals is accepted.

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

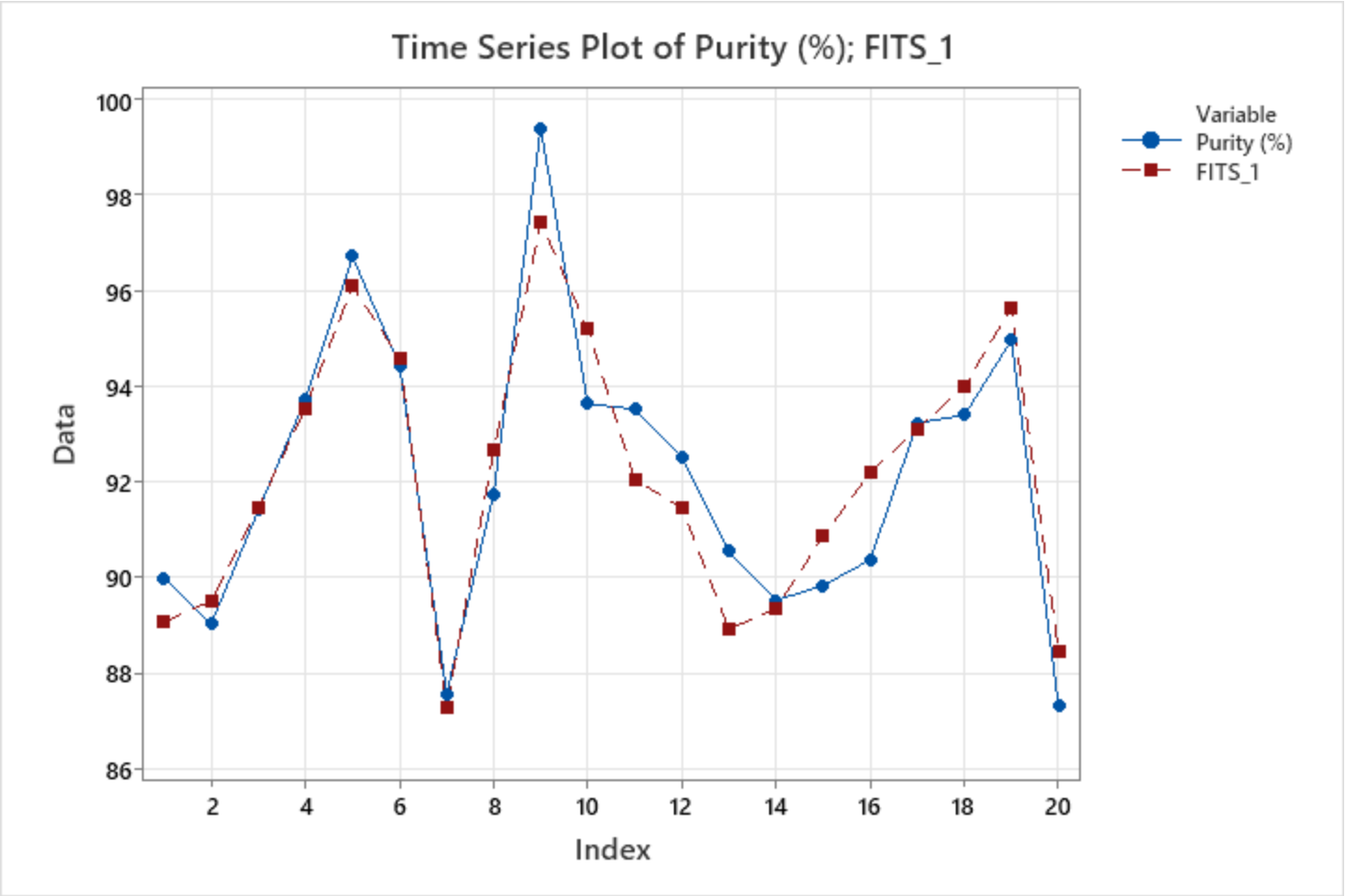
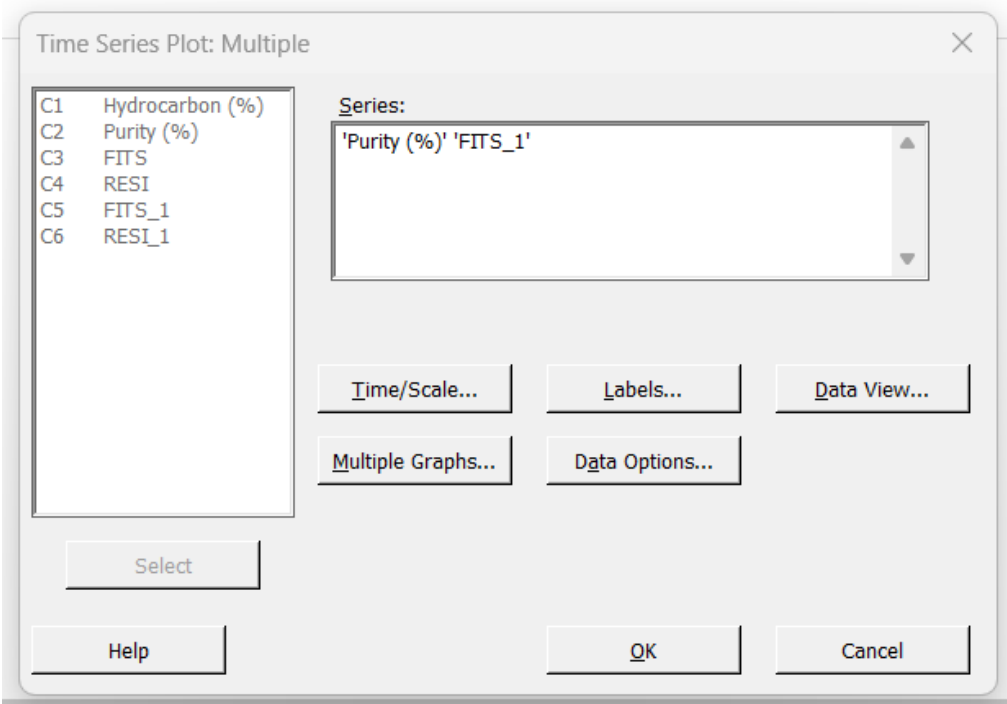


- The normal probability plot (top left) shows a normal distribution of the residuals.
- The histogram (bottom left) confirms normality.
- On top right, residuals versus fits are shown.
- The bottom right panel shows the time series plot of the residuals.

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

Plot the original data and fitted values:

Stat → Time series → Time series plot → Multiple



Regression Equation

$$\text{Purity (\%)} = 74,28 + 14,95 \text{ Hydrocarbon (\%)}$$

Exercise 1.1 Fit a simple linear regression model to the oxygen purity.

Plot the original data and fitted values:

Stat → Regression → Fitted line plot

Fitted Line Plot

C1Hydrocarbon (%)

C2Purity (%)

Response (Y):

Purity (%)

Predictor (X):

Hydrocarbon (%)

Type of Regression Model

☒ Linear

☐ Quadratic

☐ Cubic

Select

Graphs...

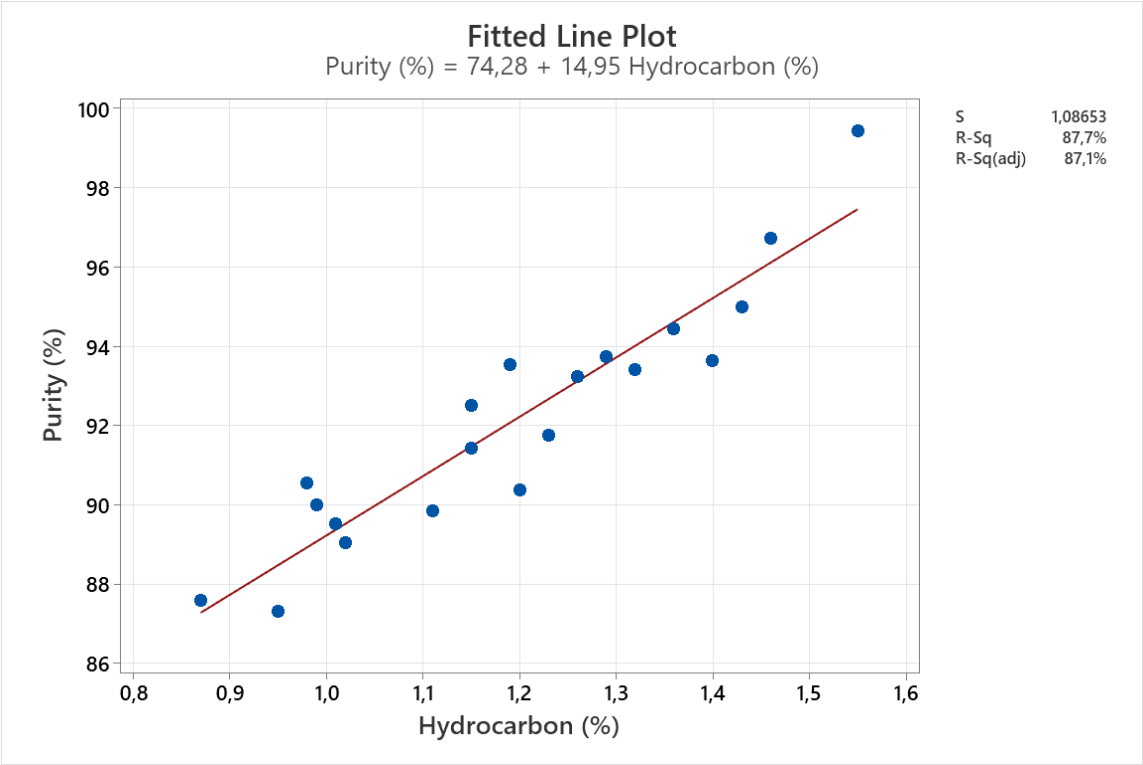
Options...

Storage...

Help

OK

Cancel



Exercise 1.2 Compute the 95% confidence interval on the slope parameter

Regression Equation

Purity (%) = 74,28 + 14,95 Hydrocarbon (%)

Confidence interval on the slope parameter

$$t_0 = \frac{\hat{\beta}_i - \beta_i}{se(\hat{\beta}_i)} \sim t_{n-p}$$
 Student - t with $n-p$ dof

p=n° predictors +1=2
n=20

$$-t_{\alpha/2, n-p} \leq \frac{\hat{\beta}_i - \beta_i}{se(\hat{\beta}_i)} \leq t_{\alpha/2, n-p}$$

dof of error term

$$\hat{\beta}_i - t_{\alpha/2, n-p} se(\hat{\beta}_i) \leq \beta_i \leq \hat{\beta}_i + t_{\alpha/2, n-p} se(\hat{\beta}_i)$$

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	74,28	1,59	46,62	0,000	
Hydrocarbon (%)	14,95	1,32	11,35	0,000	1,00

$\alpha = 5\% \Rightarrow t_{0.025, 18} = 2.1$

The standard error of the estimated slope is S = 1.32

- The confidence interval is:
 $CI = \beta_1 \pm t_{1-\alpha, n-2} * S = 14.95 \pm 2.1 * 1.32 = [12.18, 17.72]$

Exercise 1.3 Suppose that we are interested in predicting mean oxygen purity when the hydrocarbon level is 1.00%. Compute the 95% confidence interval on the mean response

Regression Equation

$$\text{Purity (\%)} = 74.28 + 14.95 \text{ Hydrocarbon (\%)}$$

Prediction at 1.00% hydrocarbon level

$$Y = 74.28 + 14.95 X + e = 74.28 + 14.95 \cdot 1 + e = 89.23 + e$$

$$\hat{Y} = 74.28 + 14.95 X = 74.28 + 14.95 \cdot 1 = 89.23$$

Let $X^* = 1\%$

Confidence interval (mean response)

$$\hat{Y} \pm t_{\frac{\alpha}{2}, n-p} \sqrt{\hat{\sigma}_\varepsilon^2 \left[\frac{1}{n} + \frac{(X^* - \bar{X})^2}{(n-1)S_X^2} \right]}$$

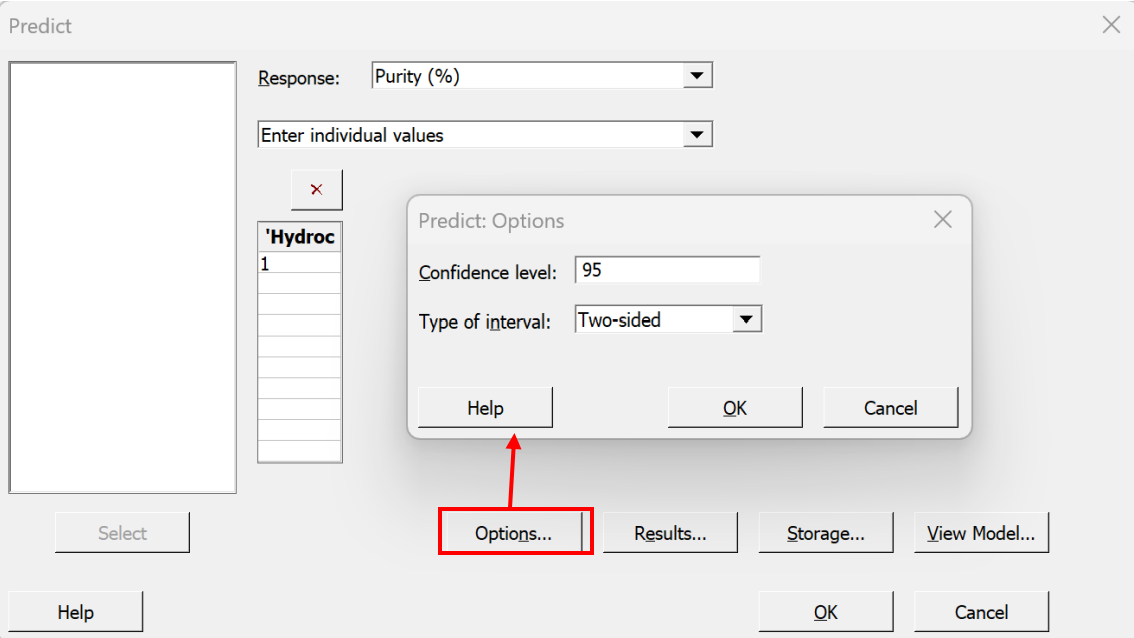
Prediction interval

$$\hat{Y} \pm t_{\frac{\alpha}{2}, n-p} \sqrt{\hat{\sigma}_\varepsilon^2 \left[1 + \frac{1}{n} + \frac{(X^* - \bar{X})^2}{(n-1)S_X^2} \right]}$$

Where $\bar{X} = 1.196$ and $S_X^2 = 0.036$, are the sample mean and sample variance of the regressor.

Exercise 1.3 Suppose that we are interested in predicting mean oxygen purity when the hydrocarbon level is 1.00%. Compute the 95% confidence interval on the mean response

Stat --> Regression --> Predict



Prediction for Purity (%)

Regression Equation

Purity (%) = 74,28 + 14,95 Hydrocarbon (%)

Settings

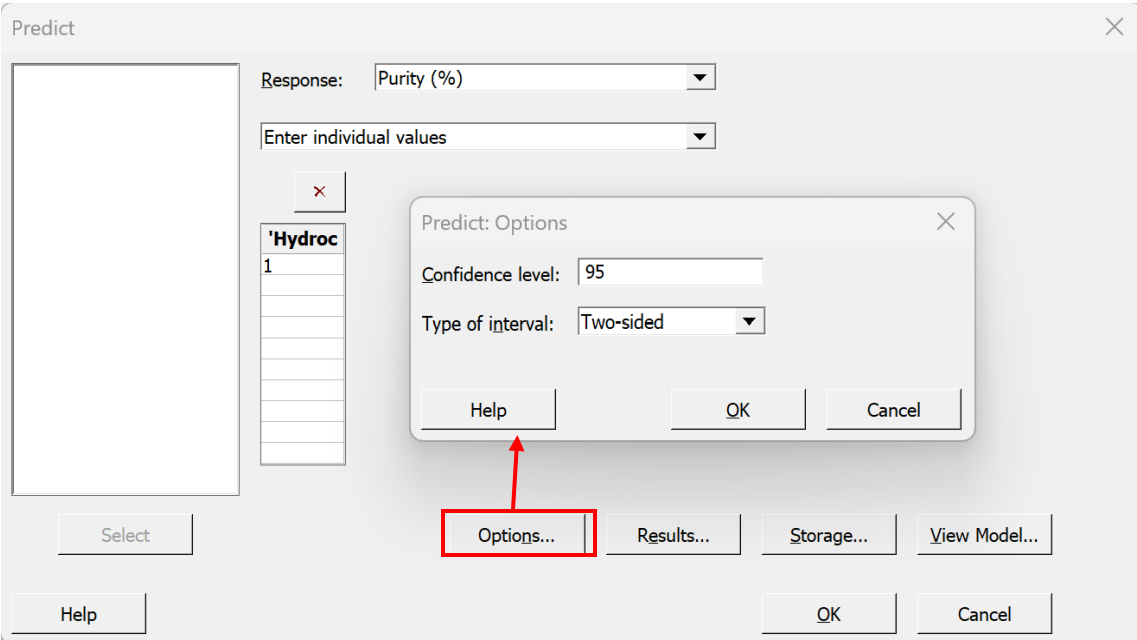
Variable	Setting
Hydrocarbon (%)	1

Prediction

Fit	SE Fit	95% CI	95% PI
89,2308	0,354450	(88,4861; 89,9755)	(86,8297; 91,6319)

Exercise 1.3 Suppose that we are interested in predicting mean oxygen purity when the hydrocarbon level is 1.00%. Compute the 95% confidence interval on the mean response

Stat --> Regression --> Predict



Prediction for Purity (%)

Regression Equation

Purity (%) = 74,28 + 14,95 Hydrocarbon (%)

Settings

Variable	Setting
Hydrocarbon (%)	1

Prediction

Fit	SE Fit	95% CI	95% PI
89,2308	0,354450	(88,4861; 89,9755)	(86,8297; 91,6319)

Exercise 1.3 Suppose that we are interested in predicting mean oxygen purity when the hydrocarbon level is 1.00%. Compute the 95% confidence interval on the mean response

Stat → Regression → Fitted line plot

Fitted Line Plot

C1 Hydrocarbon (%)
C2 Purity (%)

Response (Y): 'Purity (%)'
Predictor (X): 'Hydrocarbon (%)'

Type of Regression Model
☒ Linear ☐ Quadratic ☐ Cubic

Select

Graphs...

Options...

Storage...

Help

OK

Cancel

Fitted Line Plot: Options

Transformations
☐ Log10 of Y ☐ Display logscale for Y variable
☐ Log10 of X ☐ Display logscale for X variable

Display Options
☒ Display confidence interval
☒ Display prediction interval

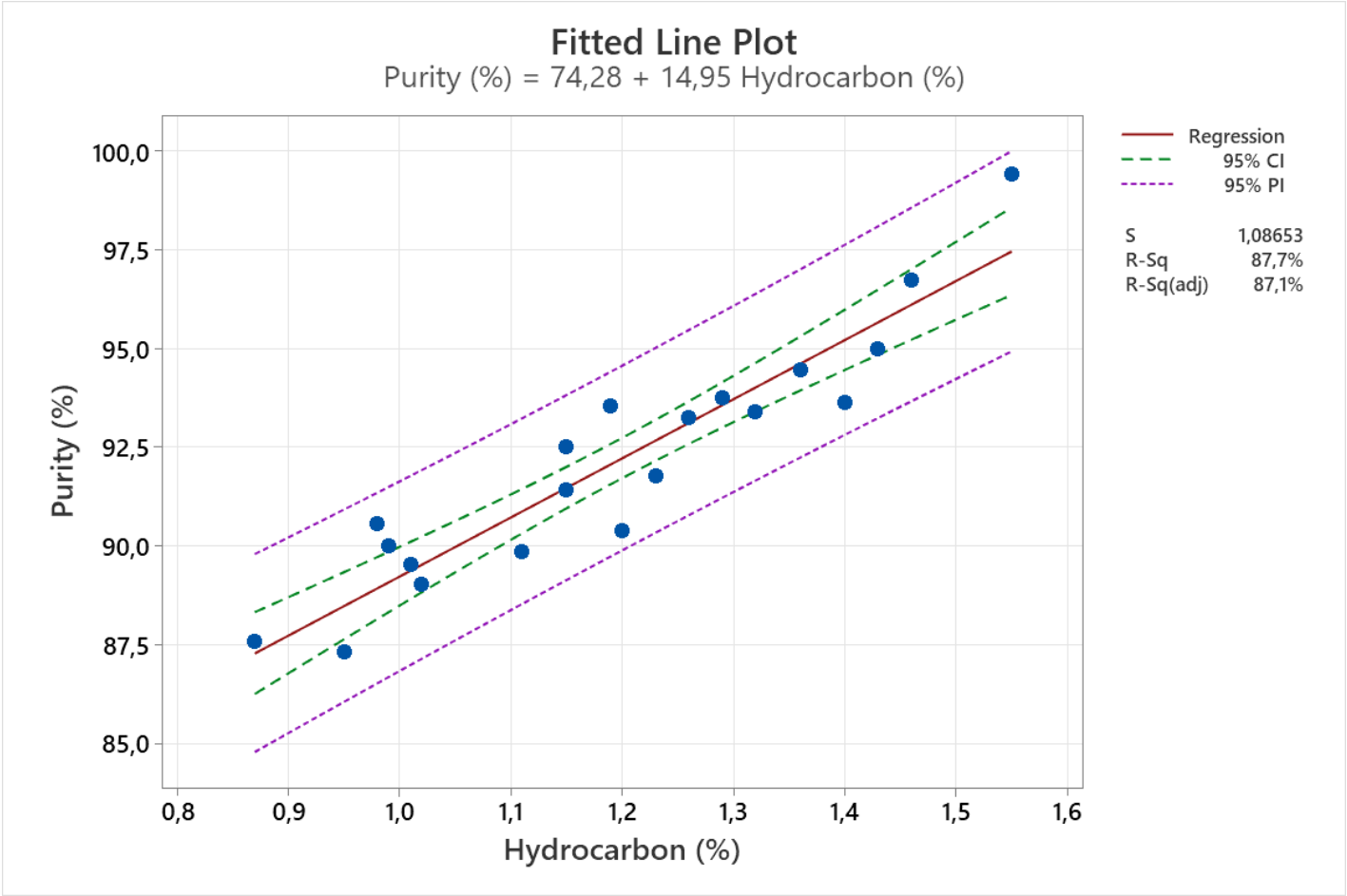
Confidence level: 95,0

Title:

Help

OK

Cancel



Exercise 2

The data in 'aircraft.csv' contains information about an industrial process for manufacturing aircraft components. The dataset includes data on the tensile Strength (kN) tested on produced parts, the Temperature (°C), Pressure (bar) and Time (minutes) of production.

1. Fit an additive multiple linear regression model to predict product strength.
2. Previous knowledge supports the hypothesis that the interaction between Temperature and Pressure could be important. Add the interaction to the model and verify its significance using a multiple linear regression model.
3. In the absence of prior knowledge regarding which interaction terms may be significant, use forward selection stepwise regression to fit a multiple linear regression model.
4. Compute the 95% confidence interval for the interaction coefficient between Temperature and Pressure in the regression model.

Exercise 2.1 Fit a multiple linear regression model to predict product strength.

Graph → Scatterplot

Scatterplot: Simple

C1 Strength

C2 Temperature

C3 Pressure

C4 Time

	Y variables	X variables
1	Strength	Temperature
2	Strength	Pressure
3	Strength	Time
4		
5		
6		

Scale...

Labels...

Data View...

Multiple Graphs...

Data Options...

Select

Help

OK

Cancel

Scatterplot: Multiple Graph

Multiple Variables | By Variables

Show Pairs of Graph Variables

☐ Overlaid on the same graph

☒ In separate panels of the same graph

☐ On separate graphs

Same Scales for Graphs

☐ Same Y

☐ Same X

Help

OK

Cancel

There is a clear linear relationship between strength (y) and time.
The relationship between strength and pressure and temperature is not clear.

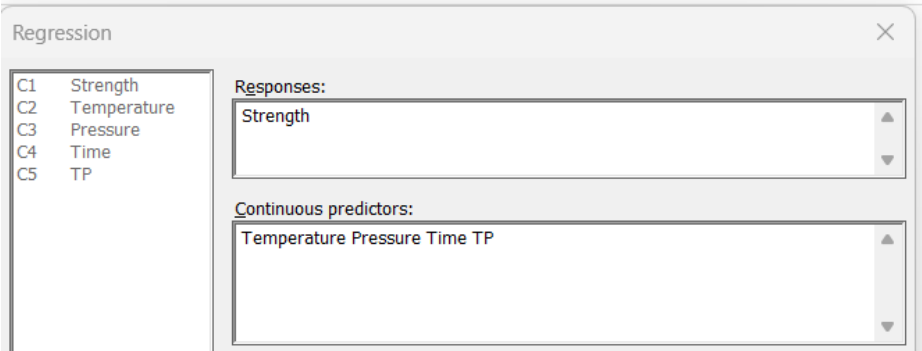
Scatterplot of Strength vs Temperature; Pressure; Time

The figure displays three scatterplots arranged in a 2x2 grid (with the bottom-right cell empty). The y-axis for all plots is 'Strength', ranging from 70 to 100. The top-left plot is 'Strength vs Temperature' (x-axis 95-115), showing a cloud of points with no clear trend. The top-right plot is 'Strength vs Pressure' (x-axis 65-80), also showing a cloud of points with no clear trend. The bottom-left plot is 'Strength vs Time' (x-axis 32-38), showing a clear positive linear relationship where strength increases from approximately 95 to 105 as time increases from 32 to 38.

20

Exercise 2.1 Fit a multiple linear regression model to predict product strength.

Stat → Regression → Regression → Fit regression model



AIRCRAFT.CSV

Regression Analysis: Strength versus Temperature; Pressure; Time

Regression Equation

Strength = 144,1 + 0,262 Temperature + 0,985 Pressure - 4,469 Time

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	144,1	24,5	5,89	0,000	
Temperature	0,262	0,150	1,75	0,092	1,12
Pressure	0,985	0,178	5,52	0,000	1,21
Time	-4,469	0,346	-12,92	0,000	1,09

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
3,18438	87,41%	85,90%	78,96%

The adj-R2 is high.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	1760,48	586,83	57,87	0,000
Temperature	1	31,04	31,04	3,06	0,092
Pressure	1	309,34	309,34	30,51	0,000
Time	1	1691,44	1691,44	166,80	0,000
Error	25	253,51	10,14		
Total	28	2013,98			

The regression is significant.
Temperature is not significant at 5%
Pressure and time are significant

Fits and Diagnostics for Unusual Observations

Obs	Strength	Fit	Resid	Std Resid
4	85,44	77,85	7,59	2,94 R
22	94,26	87,84	6,42	2,32 R

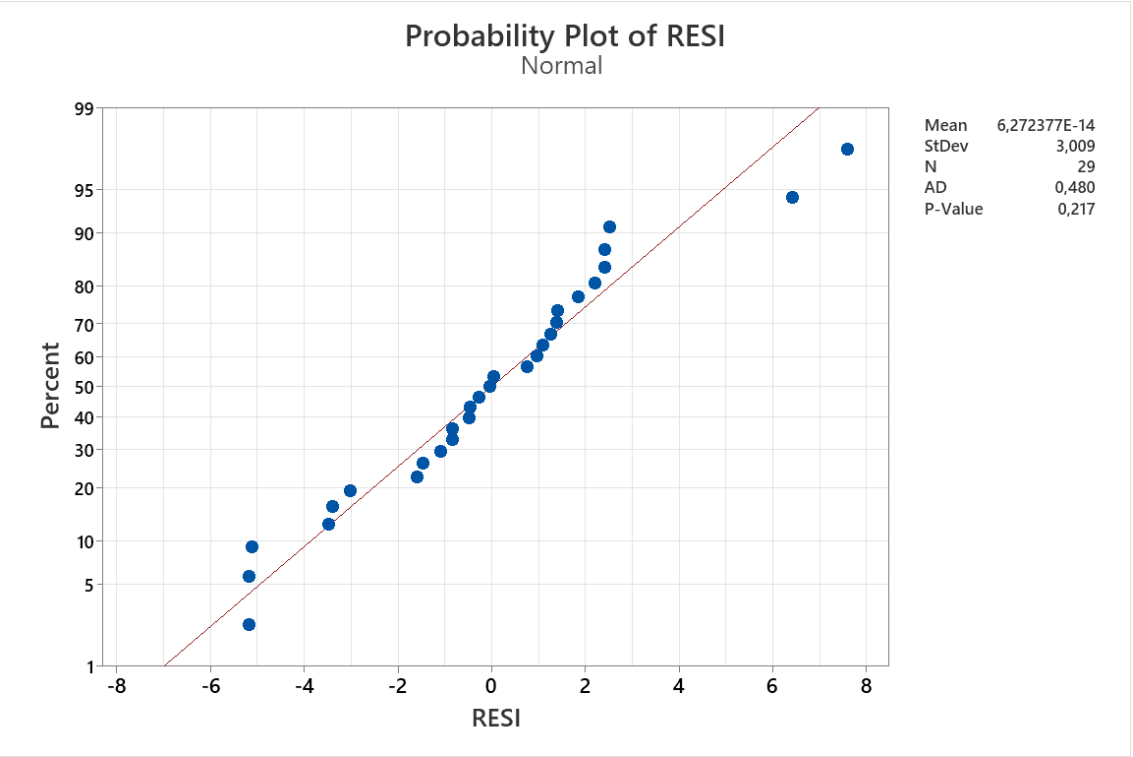
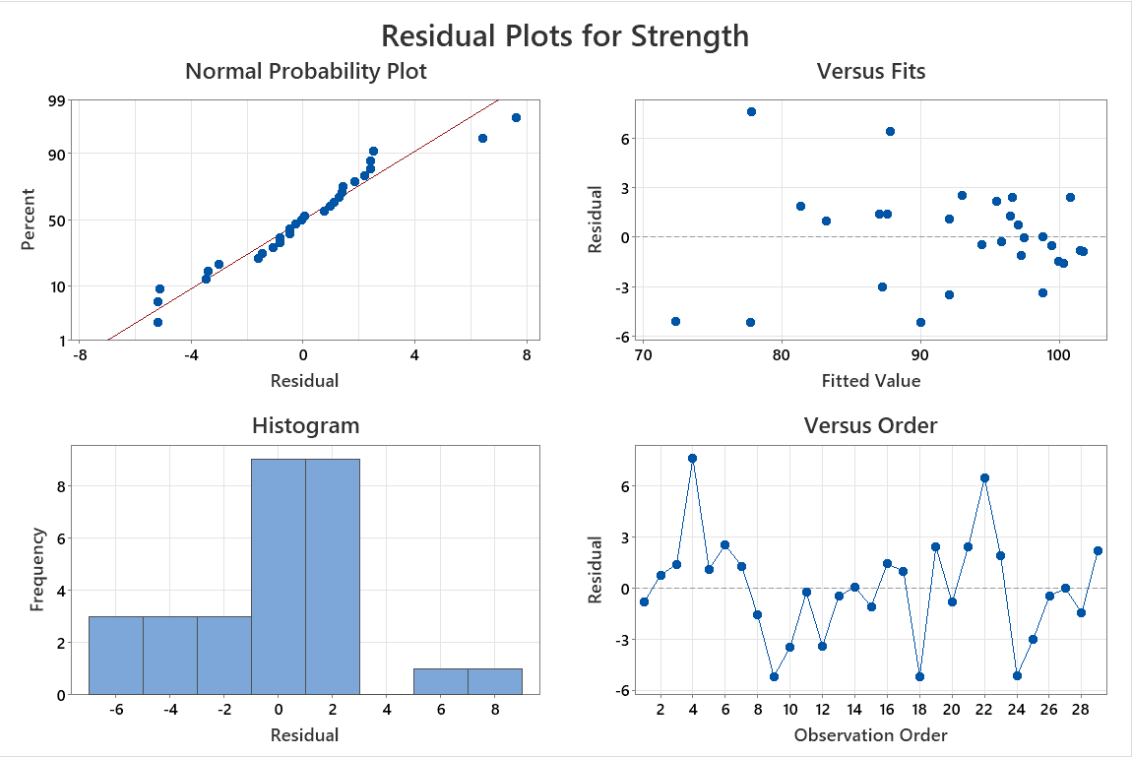
R Large residual



Minitab indicates as potential outlier these 2 obs because the standardized residuals is larger than 2.
As a rule of thumb consider an obs. as outlier when the standardized residual absolute value is higher than 3

Let's proceed by checking assumptions on residual

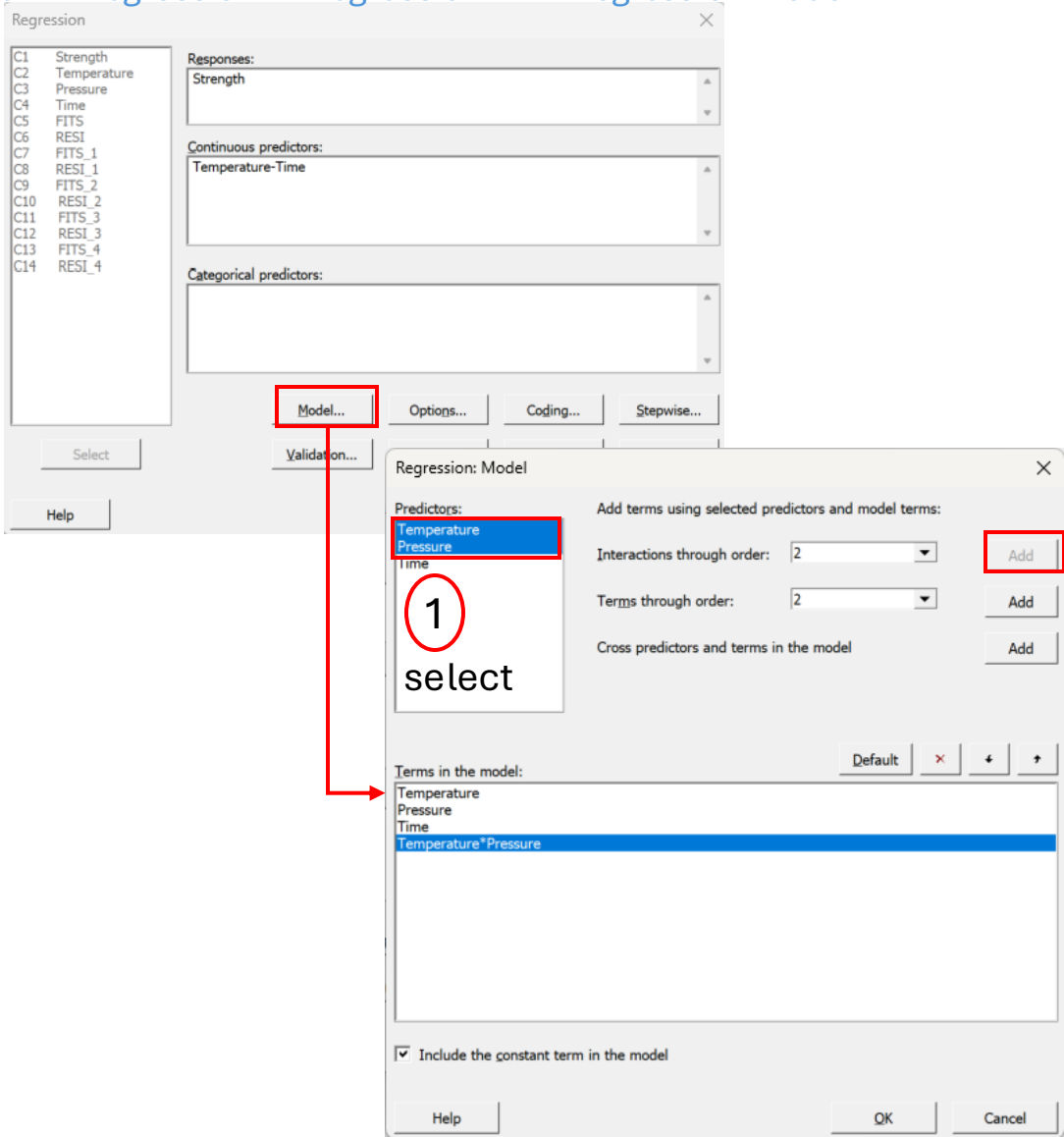
Exercise 2.1 Fit a multiple linear regression model to predict product strength.



The normality assumption cannot be rejected.
The model is validated.

Exercise 2.2 Previous knowledge supports the hypothesis that the interaction between Temperature and Pressure could be important. Add the interaction to the model and verify its significance using a multiple linear regression model.

Stat → Regression → Regression → Fit regression model



Regression Equation

$$\text{Strength} = 1064 - 8,54 \text{ Temperature} - 11,52 \text{ Pressure} - 4,774 \text{ Time} + 0,1210 \text{ Temperature*Pressure}$$

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2,82417	90,50%	88,91%	81,02%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	1822,56	455,64	57,13	0,000
Temperature	1	58,33	58,33	7,31	0,012
Pressure	1	52,62	52,62	6,60	0,017
Time	1	1712,89	1712,89	214,76	0,000
Temperature*Pressure	1	62,08	62,08	7,78	0,010
Error	24	191,42	7,98		
Total	28	2013,98			

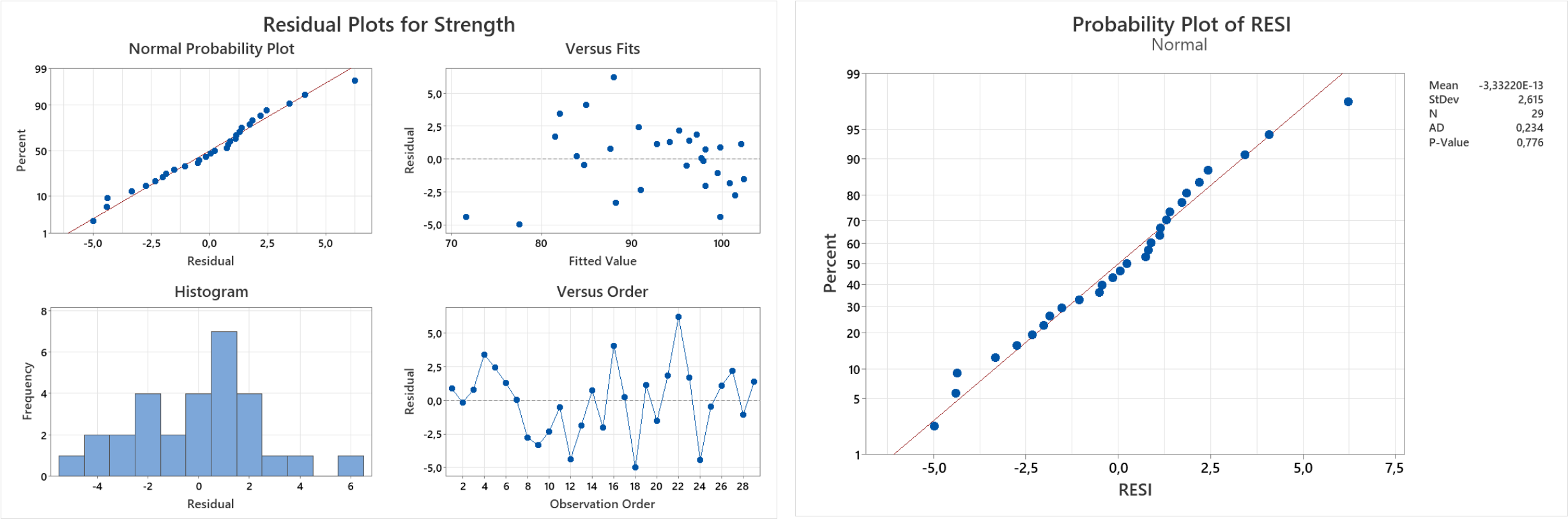
Fits and Diagnostics for Unusual Observations

Obs	Strength	Fit	Resid	Std Resid
4	85,44	82,01	3,43	1,97 X
22	94,26	88,02	6,23	2,54 R

R Large residual
X Unusual X

The interaction is significant. Let us **check the residuals**

Exercise 2.2 Previous knowledge supports the hypothesis that the interaction between Temperature and Pressure could be important. Add the interaction to the model and verify its significance using a multiple linear regression model.

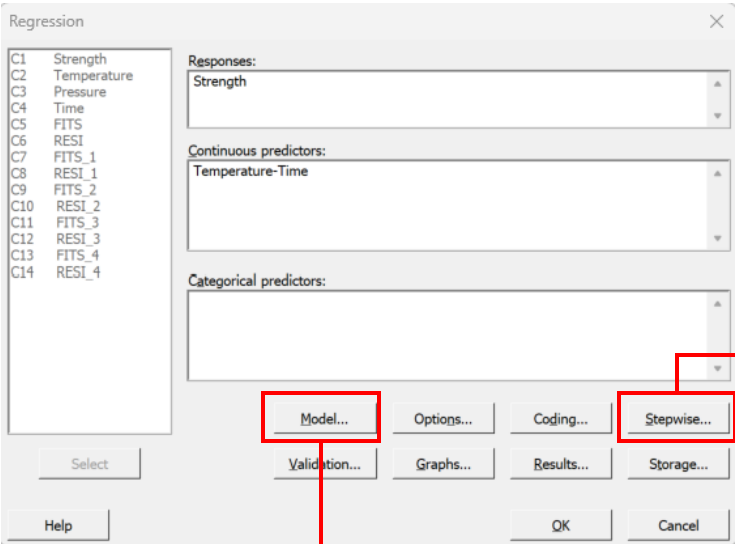


The normality assumption cannot be rejected. The model is validated and the interaction is significant.

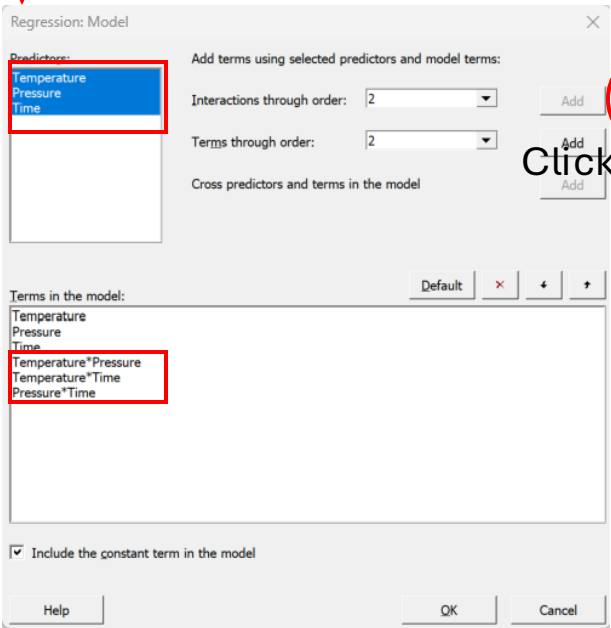
Exercise 2.2 In the absence of prior knowledge regarding which interaction terms may be significant, use forward selection stepwise regression to fit a multiple linear regression model.

Stat → Regression → Regression → Fit regression model

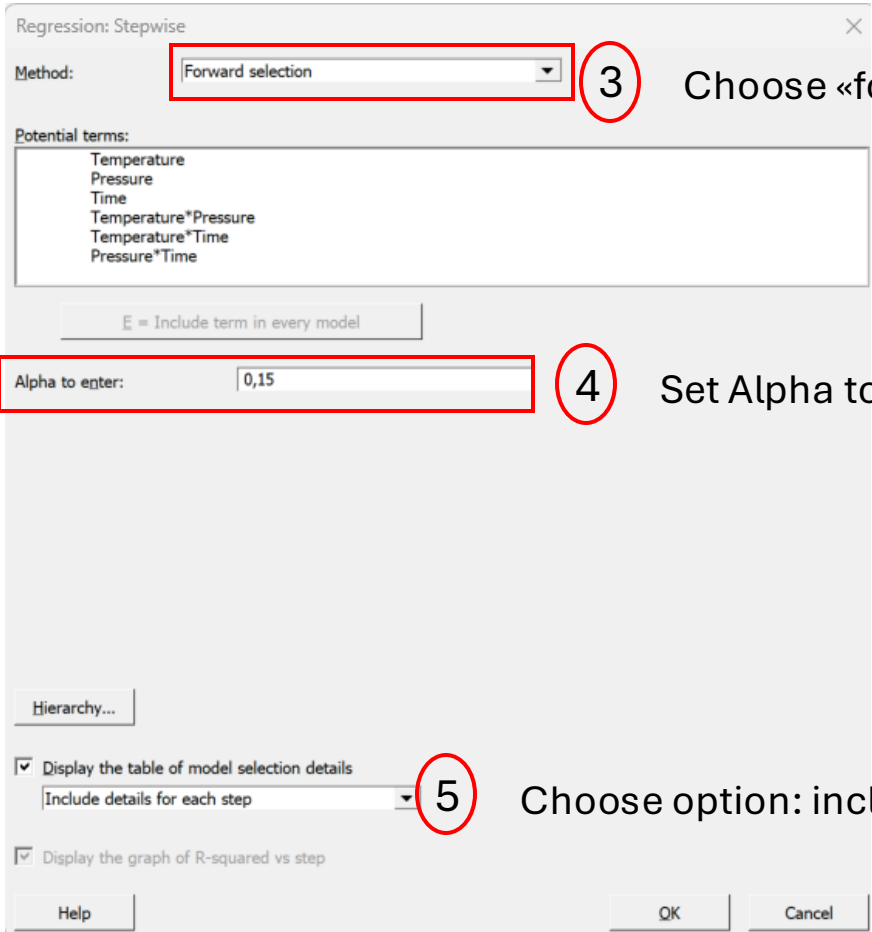
As we do not have previous knowledge on which interactions are significant, we use stepwise regression to identify the «best» model.



select 1



Click «Add»



3 Choose «forward selection»

4 Set Alpha to enter = 0,15

5 Choose option: include details for each step

Exercise 2.2 In the absence of prior knowledge regarding which interaction terms may be significant, use forward selection stepwise regression to fit a multiple linear regression model.

Forward Selection of Terms

Candidate terms: Temperature; Pressure; Time; Temperature*Pressure; Temperature*Time; Pressure*Time

	-----Step 1----		-----Step 2----		-----Step 3----	
	Coef	P	Coef	P	Coef	P
Constant	224,9		179,1		144,1	
Time	-3,963	0,000	-4,484	0,000	-4,469	0,000
Pressure			0,888	0,000	0,985	0,000
Temperature					0,262	0,092
Temperature*Pressure						
S	4,56581		3,30818		3,18438	
R-sq	72,05%		85,87%		87,41%	
R-sq(adj)	71,02%		84,78%		85,90%	
Mallows' Cp	40,53		10,13		8,51	
AICc	175,26		158,19		157,78	
BIC	178,41		161,99		162,01	

	-----Step 4----	
	Coef	P
Constant	1064	
Time	-4,774	0,000
Pressure	-11,52	0,017
Temperature	-8,54	0,012
Temperature*Pressure	0,1210	0,010

S	2,82417
R-sq	90,50%
R-sq(adj)	88,91%
Mallows' Cp	3,29
AICc	152,85
BIC	157,23

α to enter = 0,15

AIRCRAFT.CSV

Regression Analysis: Strength versus Temperature; Pressure; Time

Forward Selection of Terms

α to enter = 0,15

Regression Equation

Strength = 1064 - 8,54 Temperature - 11,52 Pressure - 4,774 Time
+ 0,1210 Temperature*Pressure

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2,82417	90,50%	88,91%	81,02%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	1822,56	455,64	57,13	0,000
Temperature	1	58,33	58,33	7,31	0,012
Pressure	1	52,62	52,62	6,60	0,017
Time	1	1712,89	1712,89	214,76	0,000
Temperature*Pressure	1	62,08	62,08	7,78	0,010
Error	24	191,42	7,98		
Total	28	2013,98			

Fits and Diagnostics for Unusual Observations

Obs	Strength	Fit	Resid	Std Resid	
4	85,44	82,01	3,43	1,97	X
22	94,26	88,02	6,23	2,54	R

R Large residual
X Unusual X

Stepwise regression finds the same model.

Exercise 2.3 Compute the 95% confidence interval for the interaction term between Temperature and Pressure in the regression model.

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	1064	331	3,22	0,004	
Temperature	-8,54	3,16	-2,70	0,012	632,33
Pressure	-11,52	4,48	-2,57	0,017	969,04
Time	-4,774	0,326	-14,65	0,000	1,23
Temp*Pres	0,1210	0,0434	2,79	0,010	1091,06

- $\alpha = 0,05$
- $n = 29$
- The confidence interval is:
$$CI = [b_{T*P} - t_{1-\alpha,n-5} * S , b_{T*P} + t_{1-\alpha,n-5} * S] = [0.032, 0.211]$$

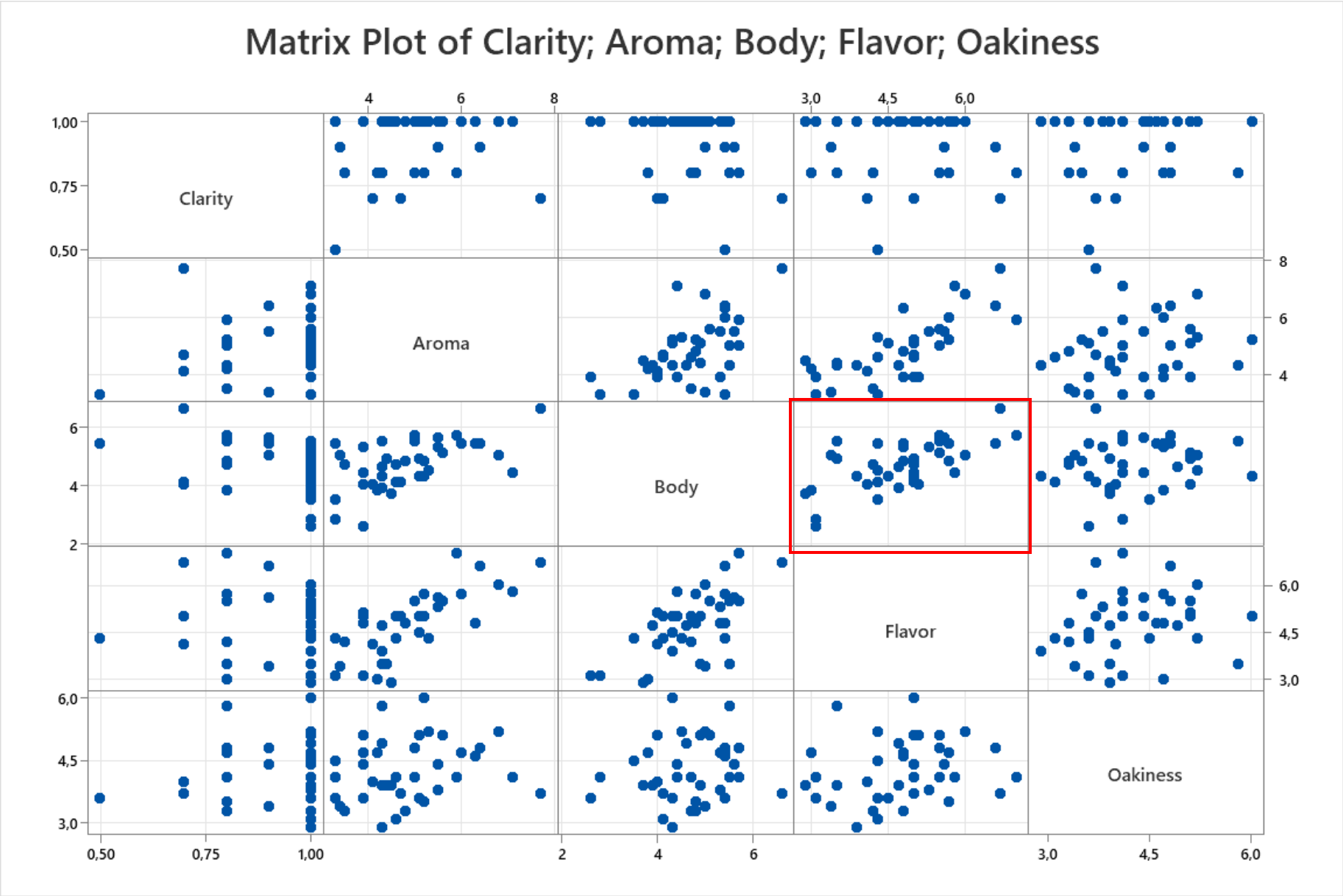
Exercise 3

'wine_quality.csv' presents data on taste-testing 38 brands of Pinot Noir wine (the data were first reported in an article by Kwan, Kowalski, and Skogenboe in the Journal of Agricultural and Food Chemistry (1979, Vol. 27)).

The response variable is $y = \text{quality}$, and we wish to find the 'best' regression equation that relates quality to the other five parameters.

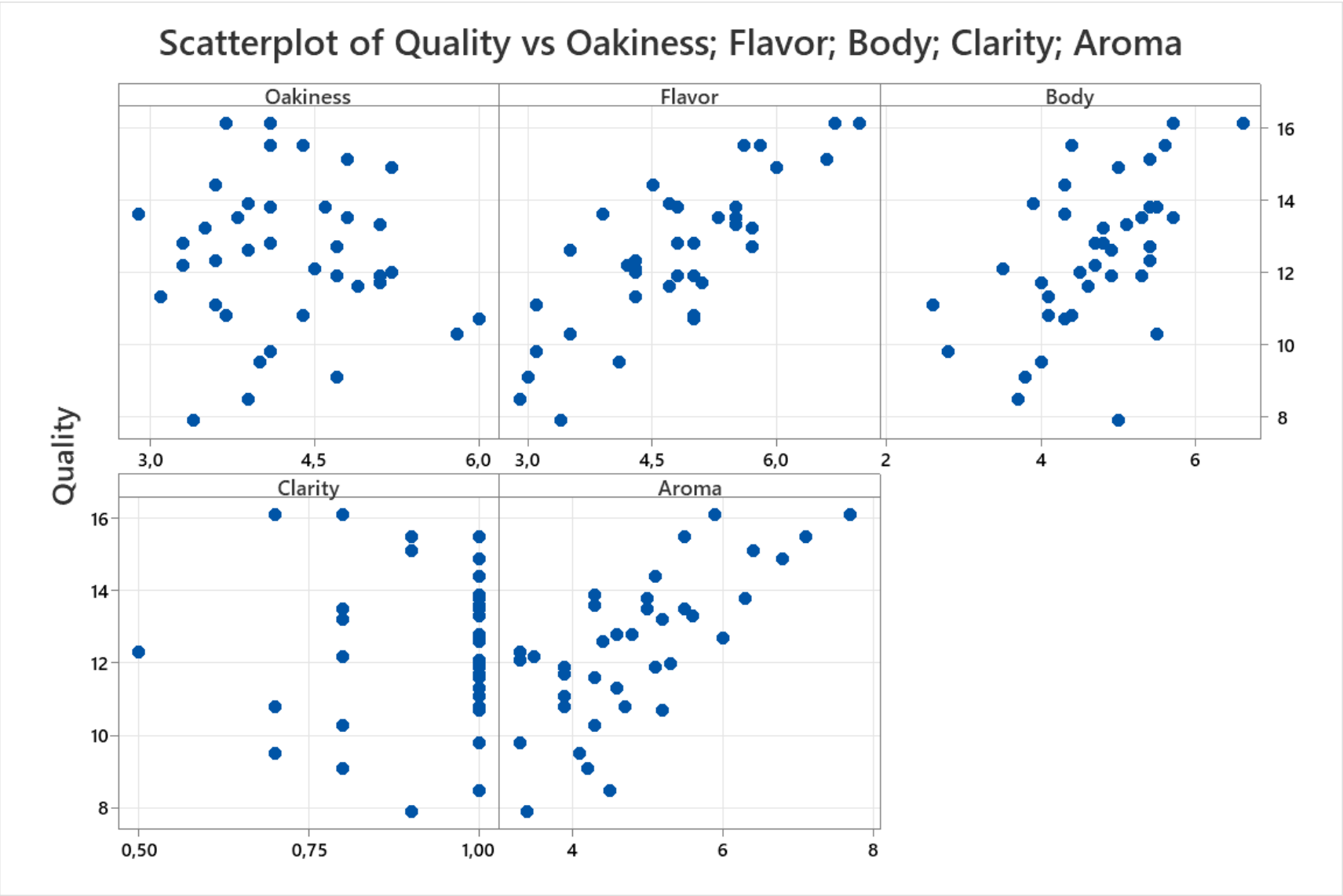
Exercise 3 Find the 'best' regression equation that relates Quality to the other five parameters

Matrix Plot of Clarity; Aroma; Body; Flavor; Oakiness



We notice that there are some indications of possible correlations between the independent variables.

Exercise 3 Find the 'best' regression equation that relates Quality to the other five parameters



The graph indicates some relationships between quality and

- Body
- Flavor
- Aroma
- Oakiness

Exercise 3 Find the 'best' regression equation that relates Quality to the other five parameters

Let's exploit the bidirectional stepwise regression algorithm to identify the significant variables and, consequently, the best regression model.

Regression Equation

Quality = 6,47 + 0,580 Aroma + 1,200 Flavor - 0,602 Oakiness

Stepwise Selection of Terms

Candidate terms: Clarity; Aroma; Body; Flavor; Oakiness

	----Step 1----		-----Step 2-----		-----Step 3-----	
	Coef	P	Coef	P	Coef	P
Constant	4,941		6,91		6,47	
Flavor	1,572	0,000	1,642	0,000	1,200	0,000
Oakiness			-0,541	0,059	-0,602	0,029
Aroma					0,580	0,034
S	1,27119		1,22423		1,16131	
R-sq	62,42%		66,11%		70,38%	
R-sq(adj)	61,37%		64,17%		67,76%	
Mallows' Cp	9,04		6,81		3,93	
AICc	130,73		129,30		126,85	
BIC	134,93		134,64		133,17	

α to enter = 0,15; α to remove = 0,15

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	6,47	1,33	4,85	0,000	
Aroma	0,580	0,262	2,21	0,034	2,21
Flavor	1,200	0,275	4,36	0,000	2,19
Oakiness	-0,602	0,264	-2,28	0,029	1,04

Regression: Stepwise

Method: Stepwise

Potential terms:

Clarity
Aroma
Body
Flavor
Oakiness

E = Include term in every modelI = Include term in the initial model

Alpha to enter: 0,15Alpha to remove: 0,15

Hierarchy...

☒ Display the table of model selection details

Include details for each step

☒ Display the graph of R-squared vs step

HelpOKCancel

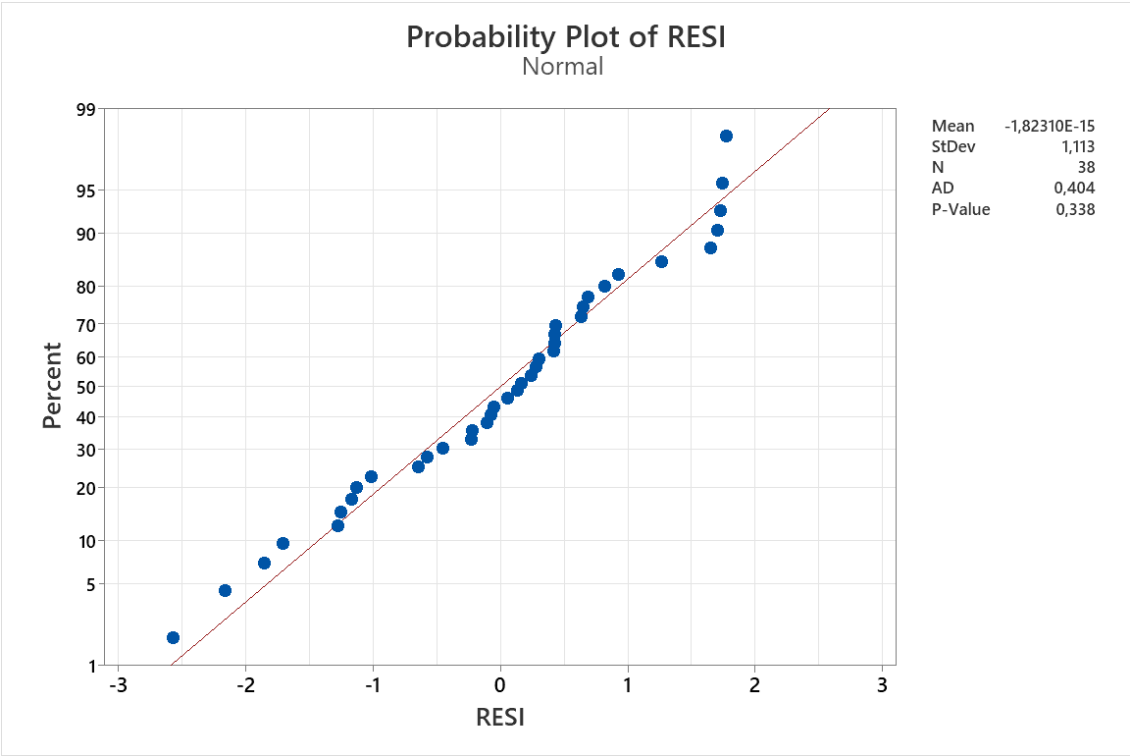
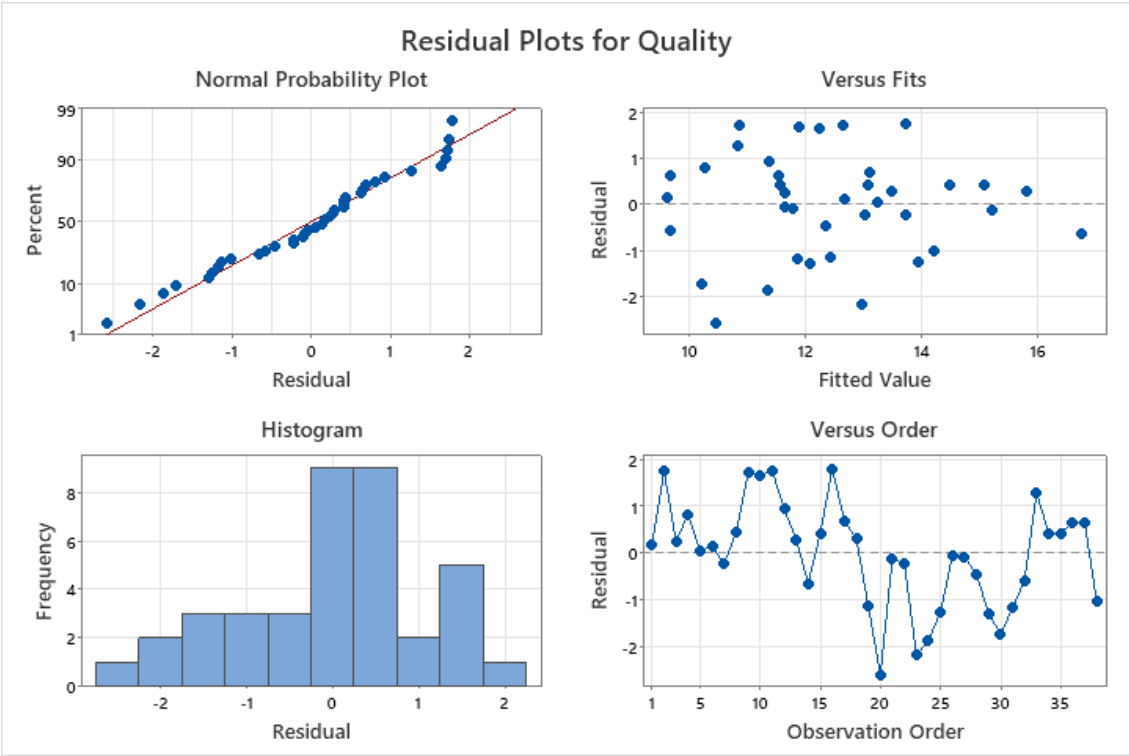
Stat → Regression → Regression → Fit regression model

Notice that the variables were entered in the order flavor (step 1), oakiness (step 2), and aroma (step 3) and that no variables were removed. No other variable could be entered, so the algorithm terminated.

Let's check assumptions on residuals.

Exercise 3 Find the 'best' regression equation that relates Quality to the other five parameters

Let's check assumptions on residuals.



The normality assumption cannot be rejected. The model is validated.

Exercise 4

In a shop floor, the head of the quality assurance department is interested in keeping under control the stability of a turning process. Every day, the cylindrical rings are produced in four temporally consecutive batches denoted as A (early morning), B (late morning), C (early afternoon), D (late afternoon). One cylindrical ring is collected and its outer diameter (cm) is measured every day in each batch. A dataset consisting of 25 consecutive sample collections is stored in 'rings.csv'.

1. Identify a suitable model.

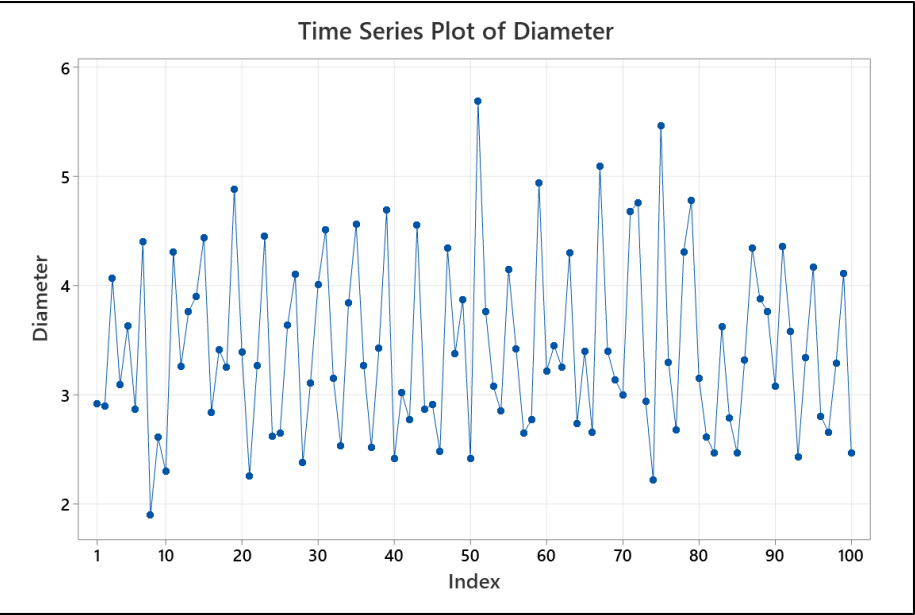
Exercise 3.1 Identify a suitable model.

1) Import the data in Minitab:

File → Open → Locate “rings.csv” and Open

2) Inspect the dataset by plotting the time series:

Graph → Time Series Plot

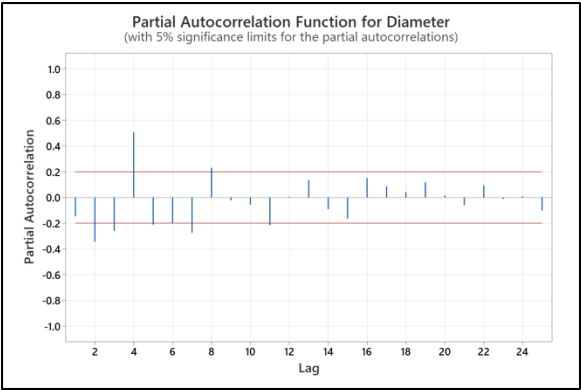
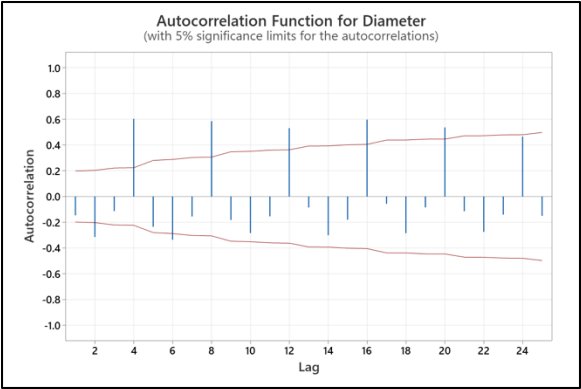


We observe a very high number of runs, a systematic variation of observations above and below the mean. This pattern is typical of negative correlated processes.

Stats → Nonparametrics → Runs test...

Number of Runs		
Observed	Expected	P-Value
57	49.00	0.094

Stats → Time Series → ACF/PACF...



Runs test would lead to not reject the randomness assumption at 95% confidence, but the time series exhibits a clear pattern, likely because of a **batch effect**.

Exercise 3.1 Identify a suitable model.

3) Inspect the possible batch effect:

Let’s create a column for Day subscripts:

Calc → Make Patterned Data → Simple Set of Numbers

Simple Set of Numbers

×

Store patterned data in: C2

From first value: 1

To last value: 25

In steps of: 1

Number of times to list each value: 4

Number of times to list the sequence: 1

Select

Help

OK

Cancel

Let’s create a column for Batch subscripts:

Calc → Make Patterned Data → Simple Set of Numbers

Simple Set of Numbers

×

Store patterned data in: C3

From first value: 1

To last value: 4

In steps of: 1

Number of times to list each value: 1

Number of times to list the sequence: 25

Select

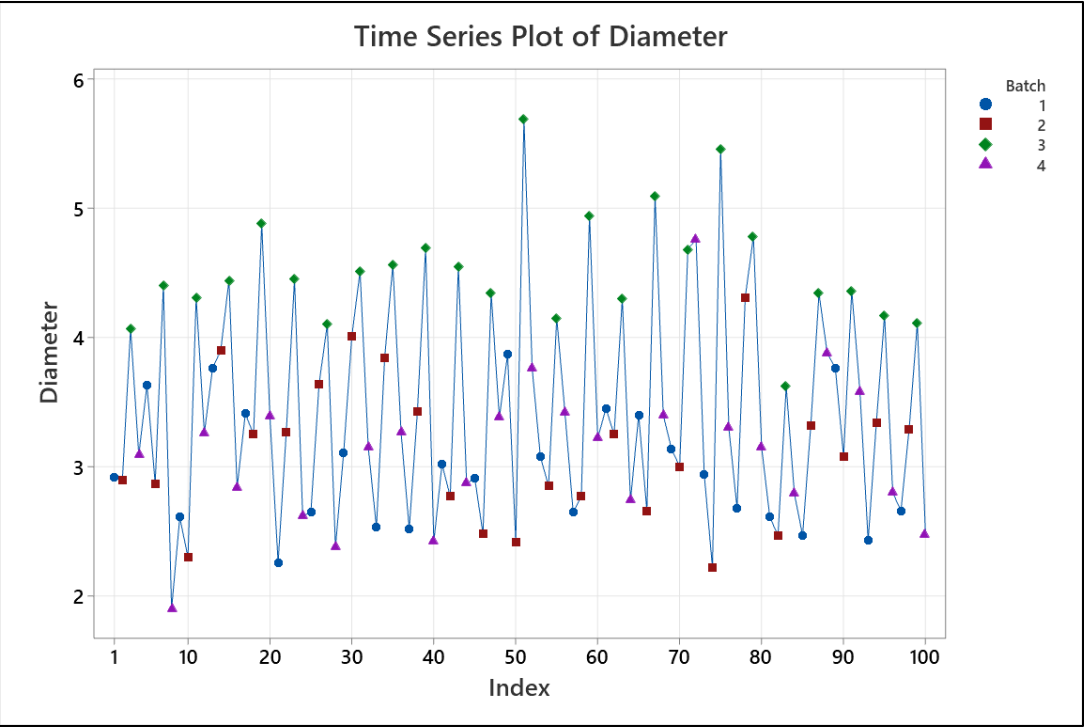
Help

OK

Cancel

Update the time series plot of Diameter:

Double click on the plot → Double click on one observation point → Go to groups tab → Insert C3/Batch as grouping variable.



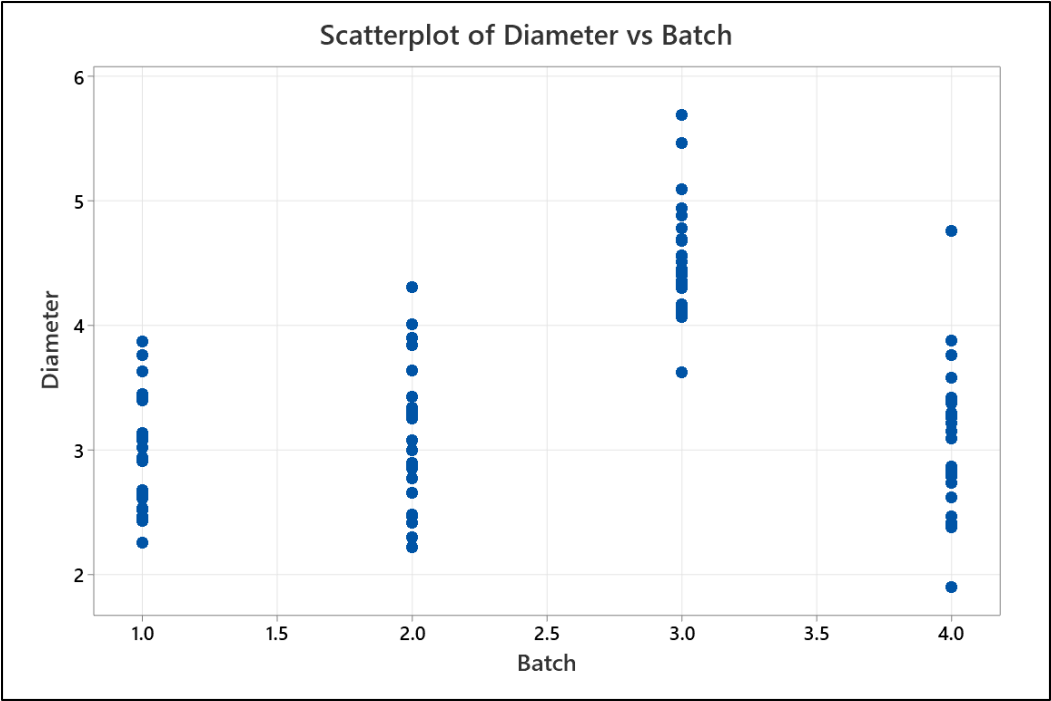
Batch 3 yields systematically a larger diameter than the other batches.

Exercise 3.1 Identify a suitable model.

3) Inspect the possible batch effect:

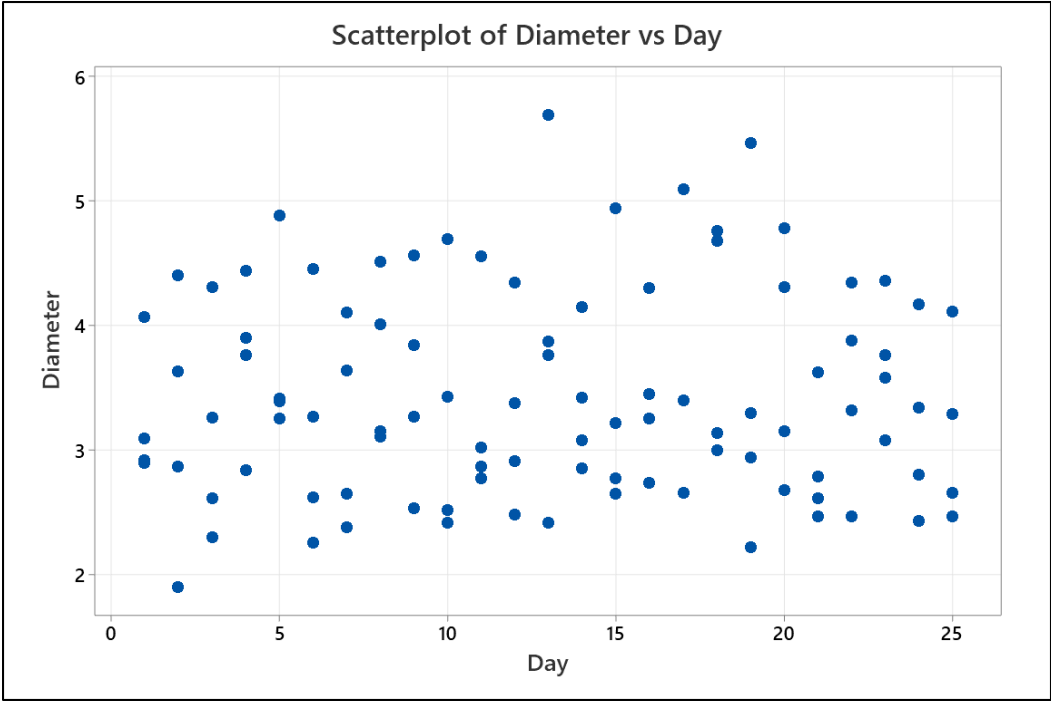
Let's create a scatterplot of Diameter vs Batch:

Graph → Scatterplot → Simple → Y: Diameter, X: Batch



What about Diameter vs Day?:

Graph → Scatterplot → Simple → Y: Diameter, X: Day



Exercise 3.1 Identify a suitable model.

4) Let’s model the batch effect – Strategy 1:

Since batch C exhibits a systematically larger diameter, we create a new dummy variable = 0 for batches A, B, D and = 1 for batch C.

Let’s create a column for Day subscripts:

Calc → Make Patterned Data → Arbitrary Set of Numbers...

Arbitrary Set of Numbers

Store patterned data in: C4

Arbitrary set of numbers:
0 0 1 0

Number of times to list each value: 1

Number of times to list the sequence: 25

Select

Help

OK

Cancel

Let’s fit a regression model:

Stat → Regression → Regression → Fit regression model...

Regression

C1 Diameter
C2 Day
C3 Batch
C4 Dummy
C5 FITS
C6 RES1
C7 COEF

Responses:
Diameter

Continuous predictors:
Dummy

Categorical predictors:

Model... Options... Coding... Stepwise...

Select

Validation... Graphs... Results... Storage...

Help

OK

Cancel

Regression Equation

Diameter = 3.0660 + 1.454 Dummy

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.511952	60.67%	60.27%	59.17%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	3.0660	0.0591	51.86	0.000	
Dummy	1.454	0.118	12.29	0.000	1.00

Exercise 3.1 Identify a suitable model.

4) Let’s model the batch effect – Strategy 1:

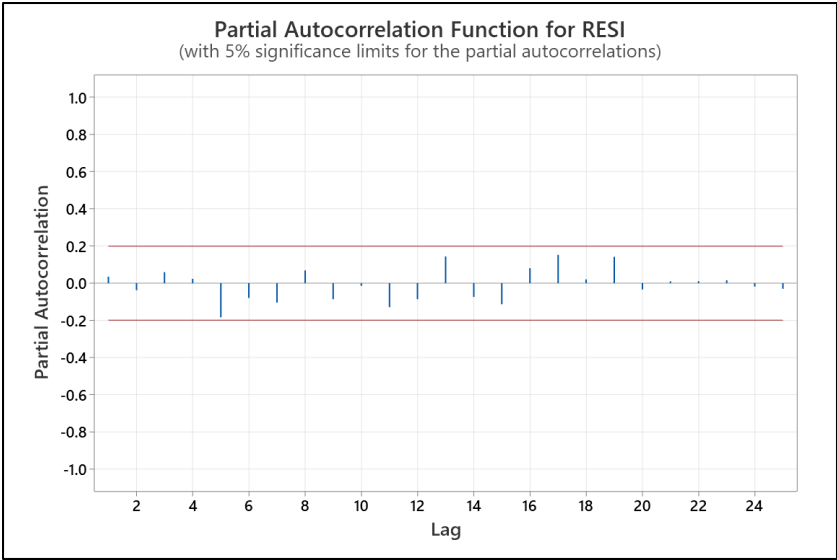
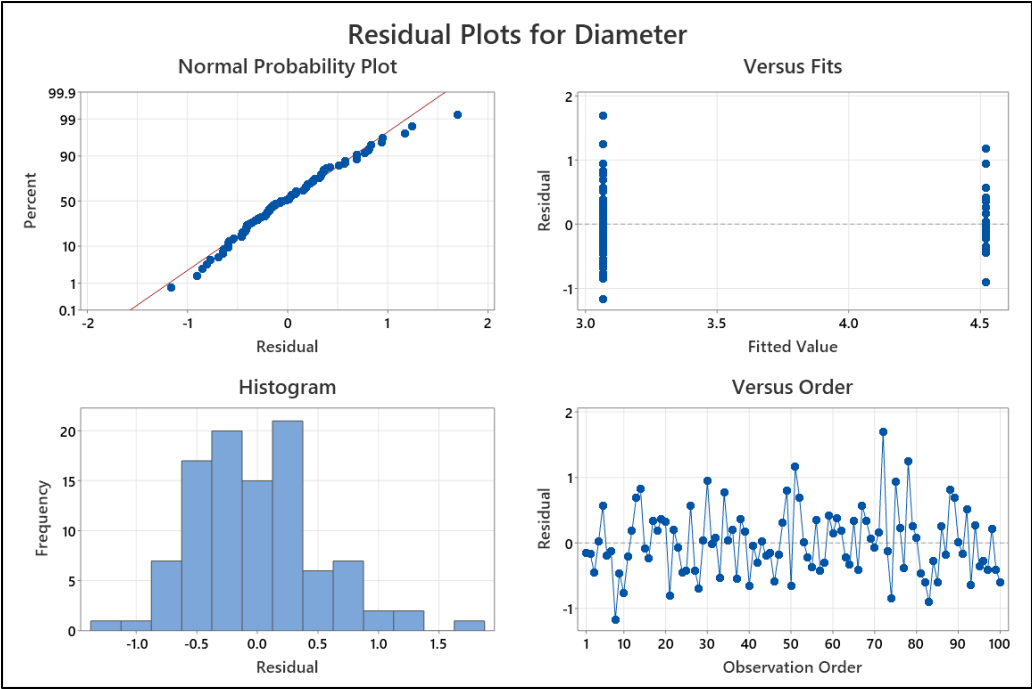
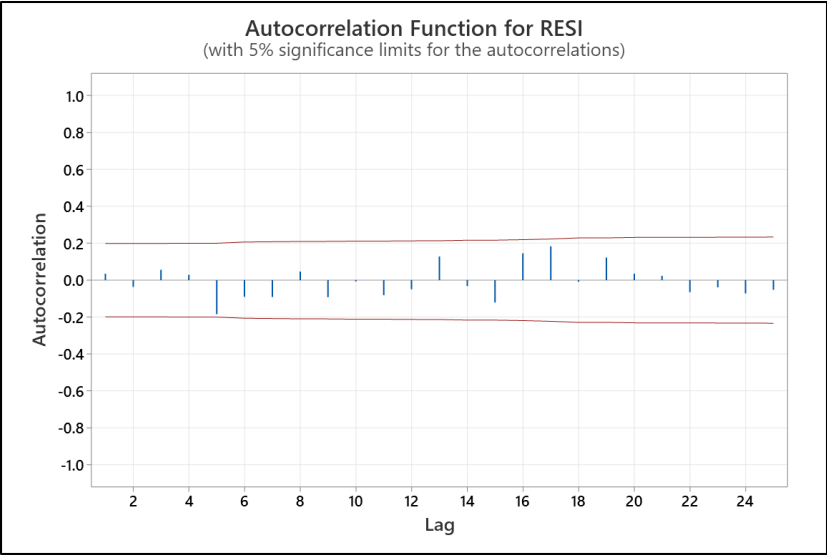
Check assumptions on residuals.

Runs test

Number of Runs		
Observed	Expected	P-Value
49	50.92	0.699

Normality test

Mean	-1.00808E-15
StDev	0.5094
N	100
RJ	0.989
P-Value	>0.100



Exercise 3.1 Identify a suitable model.

5) Let’s model the batch effect – Strategy 2:

Let's use a set of categorical dummy variables (Dummy_Batch1, Dummy_Batch2, Dummy_Batch3, Dummy_Batch4)

Calc → Make Patterned Data → Arbitrary Set of Numbers...

Arbitrary Set of Numbers

Store patterned data in: C8

Arbitrary set of numbers: 1 0 0 0

Number of times to list each value: 1

Number of times to list the sequence: 25

Select OK Cancel

Arbitrary Set of Numbers

Store patterned data in: C8

Arbitrary set of numbers: 0 1 0 0

Number of times to list each value: 1

Number of times to list the sequence: 25

Select OK Cancel

Arbitrary Set of Numbers

Store patterned data in: C9

Arbitrary set of numbers: 0 0 1 0

Number of times to list each value: 1

Number of times to list the sequence: 25

Select OK Cancel

Arbitrary Set of Numbers

Store patterned data in: C11

Arbitrary set of numbers: 0 0 0 1

Number of times to list each value: 1

Number of times to list the sequence: 25

Select OK Cancel

C8	C9	C10	C11
Batch1	Batch2	Batch3	Batch4
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

Let’s fit a regression model. Since we have 4 batches, it is appropriate to use 3 dummy variables as regressors. The fourth dummy variable can be inferred from the other three (Dummy_Batch3 = 1 - Dummy_Batch1 - Dummy_Batch2 - Dummy_Batch4). Let's leave out Dummy_Batch3 and use the other three dummy variables as regressors.

Stat → Regression → Regression → Fit regression model...

Regression

Responses: Diameter

Continuous predictors: Batch1 Batch2 Batch4

Categorical predictors:

Model... Options... Coding... Stepwise... Validation... Graphs... Results... Storage... Select OK Cancel

Regression Equation

Diameter = 4.520 - 1.541 Batch1 - 1.414 Batch2 - 1.406 Batch4

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.514370	61.11%	59.89%	57.80%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	4.520	0.103	43.93	0.000	
Batch1	-1.541	0.145	-10.59	0.000	1.50
Batch2	-1.414	0.145	-9.72	0.000	1.50
Batch4	-1.406	0.145	-9.66	0.000	1.50

Exercise 3.1 Identify a suitable model.

5) Let’s model the batch effect – Strategy 2:

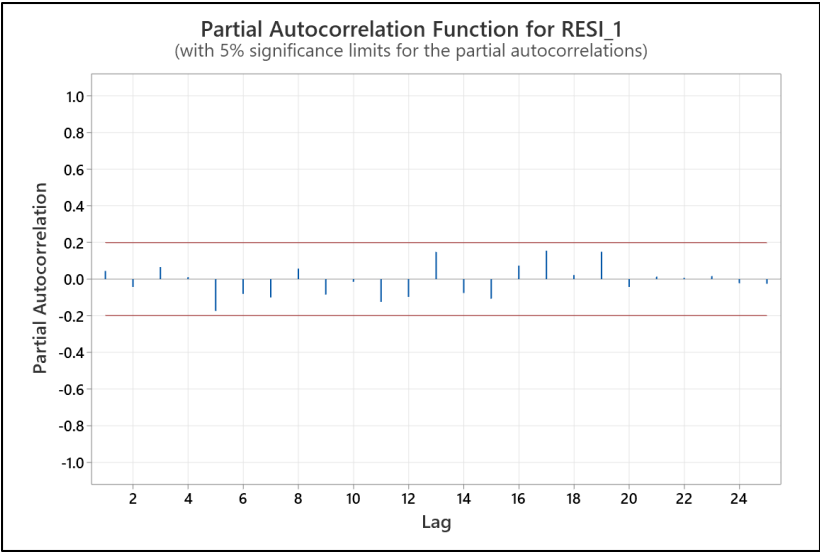
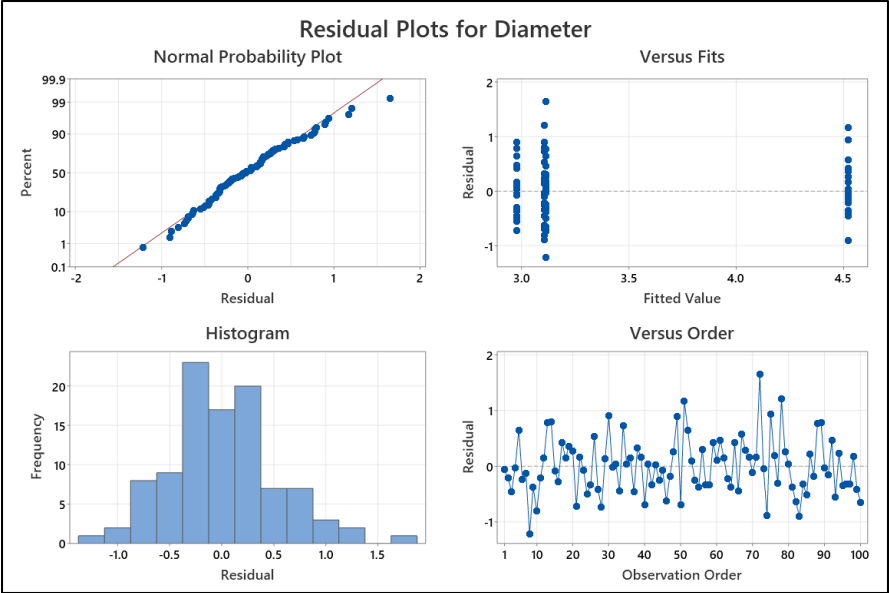
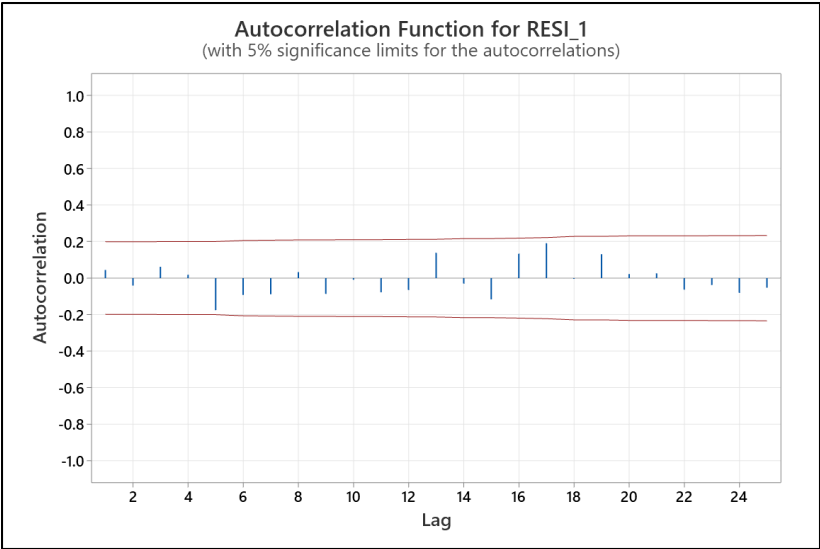
Let’s check assumptions on residuals.

Runs test

Number of Runs		
Observed	Expected	P-Value
51	50.82	0.971

Normality test

Mean	-1.23013E-15
StDev	0.5065
N	100
RJ	0.990
P-Value	>0.100



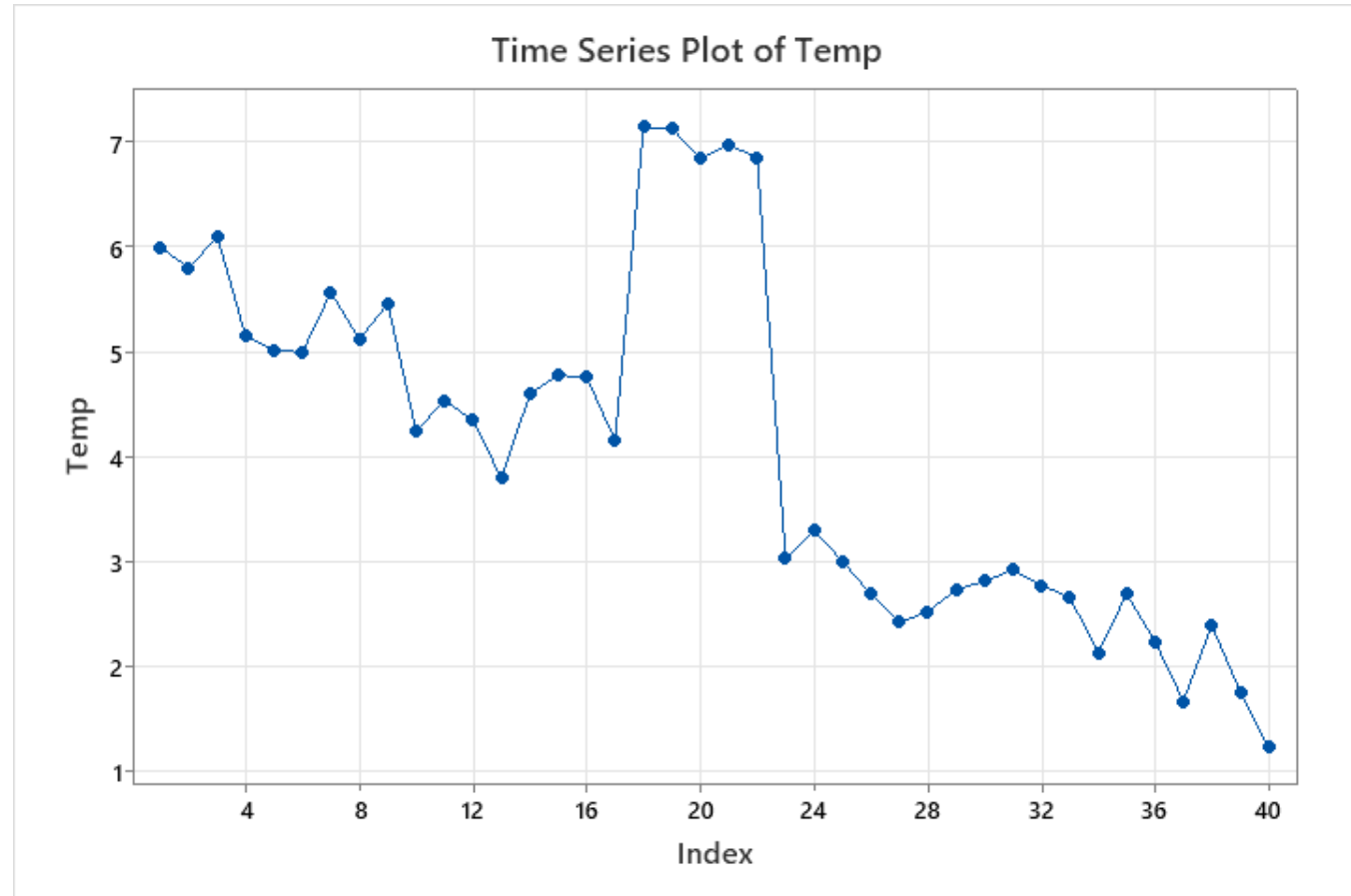
Both strategies lead to adequate models. The second model is similar to the first one, with an R-squared adjusted of 0.599 (against the previous of 0.603).

EXERCISE 5

A scientific instrument installed on the International Space Station is kept at a low temperature by means of a liquid cooling system. The average temperature measured every day for 40 consecutive days are reported in 'temperatures.csv'.

From day 18 to day 22 the instrument was used for some extra experiments, whereas normal operating conditions were applied in all the remaining days.

1. Identify a suitable model.



Point 1

Identify a suitable model.

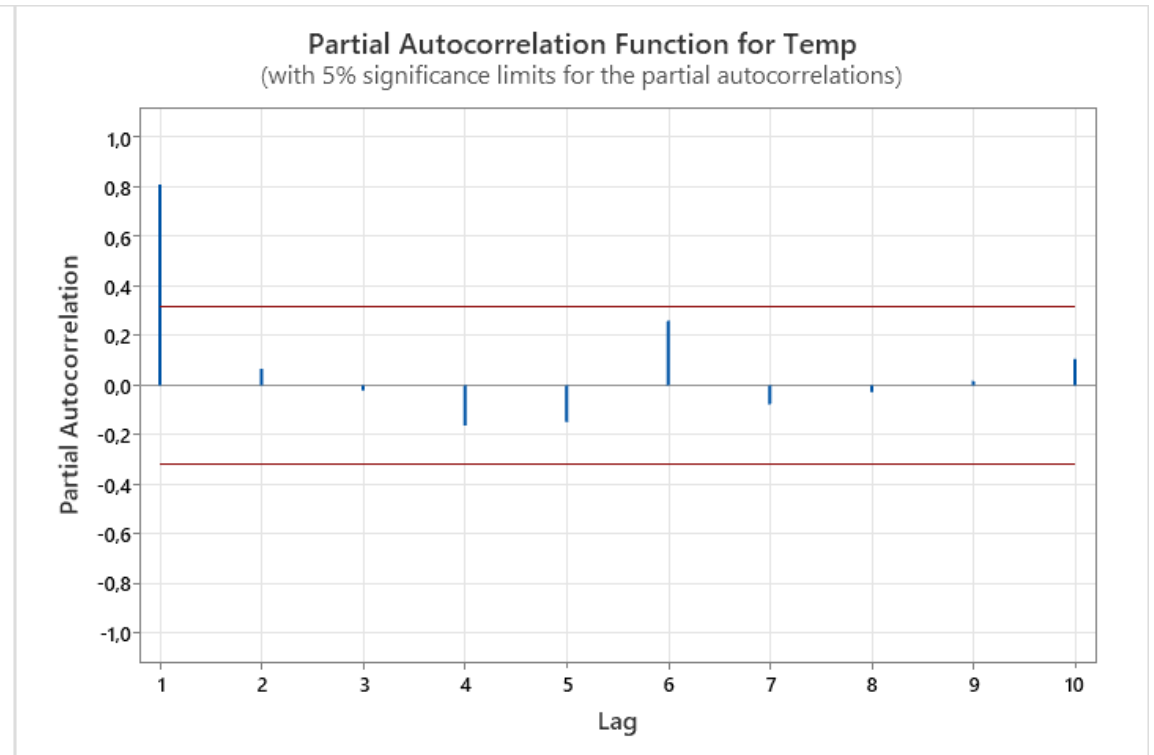
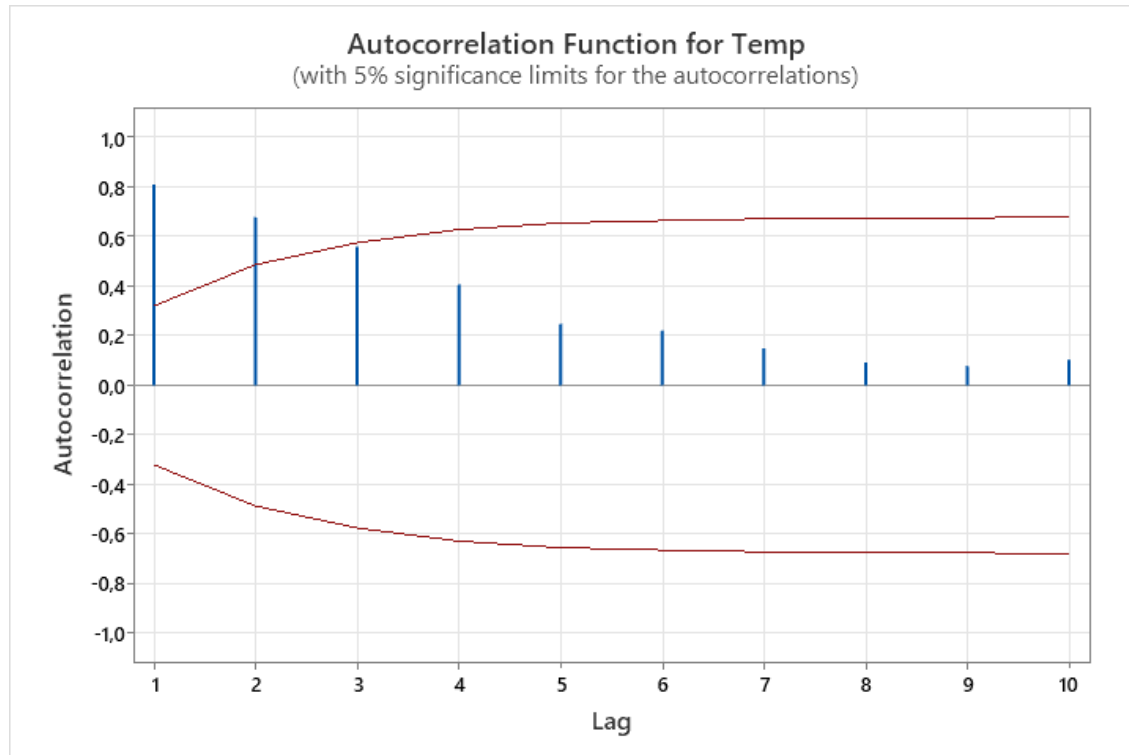
Let's verify the independence assumption with
Runs test and ACF/PACF:

Runs test

Null hypothesis H_0 : The order of the data is random

Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
4	20,95	0,000



There is a **trend**, with a **jump** in the middle for days 18-22.

Point 1

Identify a suitable model.

Let's try to fit a model with a **dummy variable** that is non-null just for days 18-22.
Create a model in the form:

$$X_t = \beta_0 + \beta_1 day + \beta_2 dummy + \epsilon_t$$

Stat → Regresssion → Regression → Fit regression model

The screenshot shows the Minitab Regression dialog box. On the left, a list of variables includes C1 Temp, C2 Day, and C3 Dummy. The 'Responses' section contains 'Temp'. The 'Continuous predictors' section contains 'Day'. The 'Categorical predictors' section contains 'Dummy'. At the bottom, there are buttons for 'Model...', 'Options...', 'Coding...', 'Stepwise...', 'Validation...', 'Graphs...', 'Results...', 'Storage...', 'Select', 'Help', 'OK', and 'Cancel'. The 'Model...' button is highlighted with a red box.

	Temp	Day	Dummy
1	6,00	1	0
2	5,80	2	0
3	6,10	3	0
4	5,15	4	0
5	5,02	5	0
6	5,00	6	0
7	5,57	7	0
8	5,12	8	0
9	5,46	9	0
10	4,25	10	0
11	4,54	11	0
12	4,35	12	0
13	3,80	13	0
14	4,60	14	0
15	4,78	15	0
16	4,76	16	0
17	4,16	17	0
18	7,14	18	1
19	7,13	19	1
20	6,84	20	1
21	6,97	21	1
22	6,85	22	1
23	3,04	23	0
24	3,31	24	0
25	3,01	25	0

Point 1

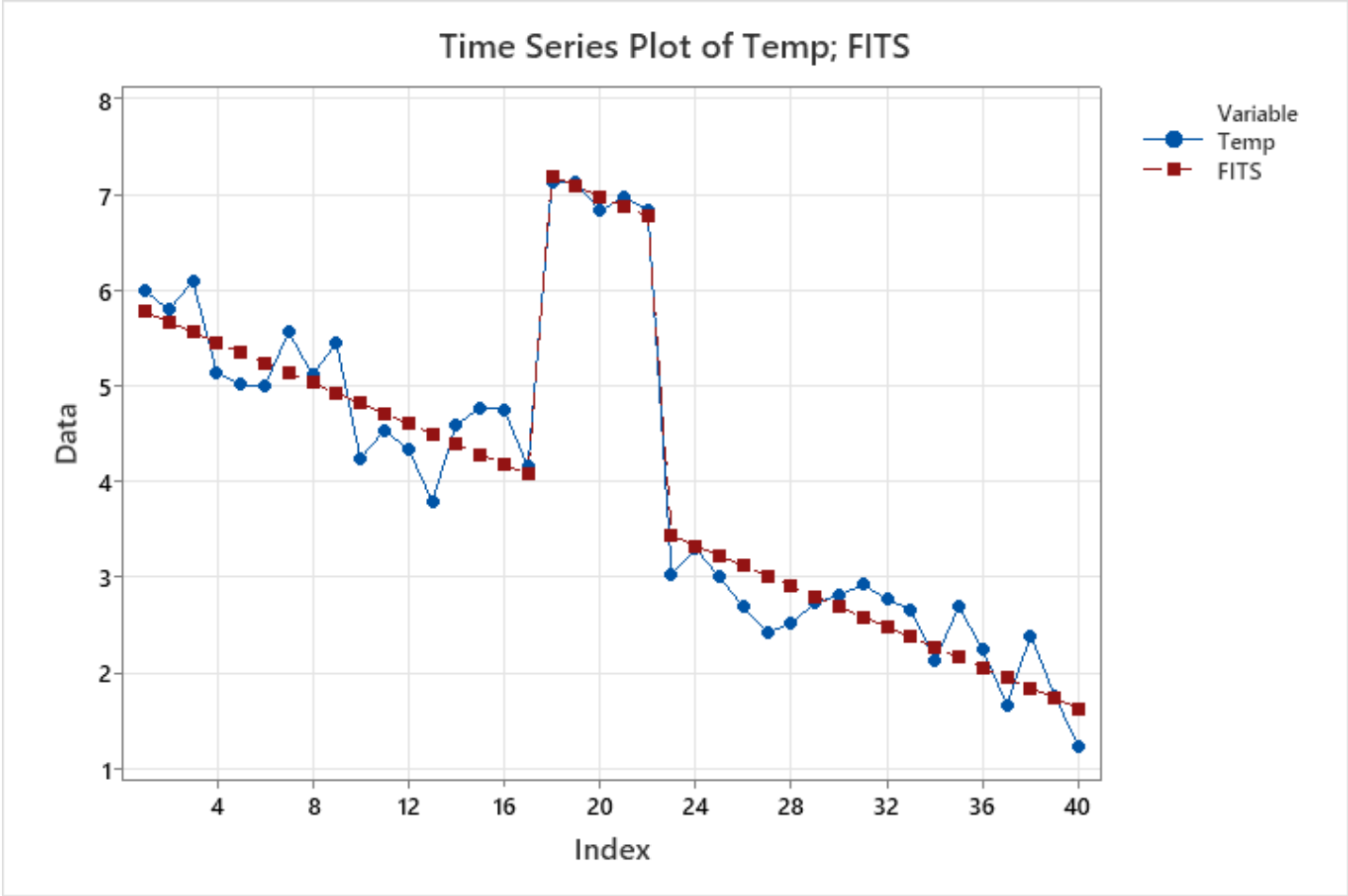
Identify a suitable model.

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	5,886	0,117	50,11	0,000	
Day	-0,10615	0,00490	-21,68	0,000	1,00
Dummy					
1	3,222	0,171	18,85	0,000	1,00

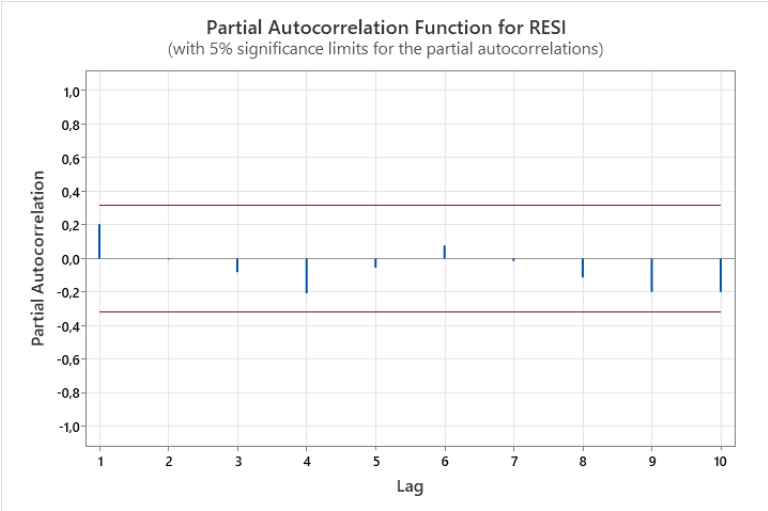
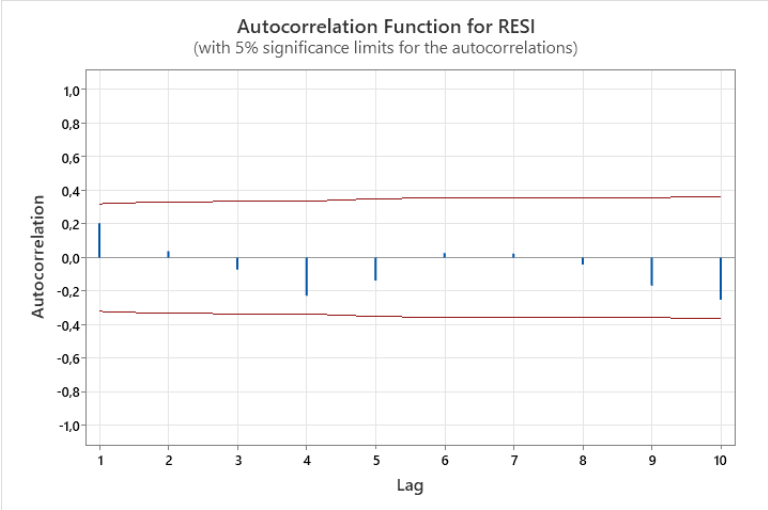
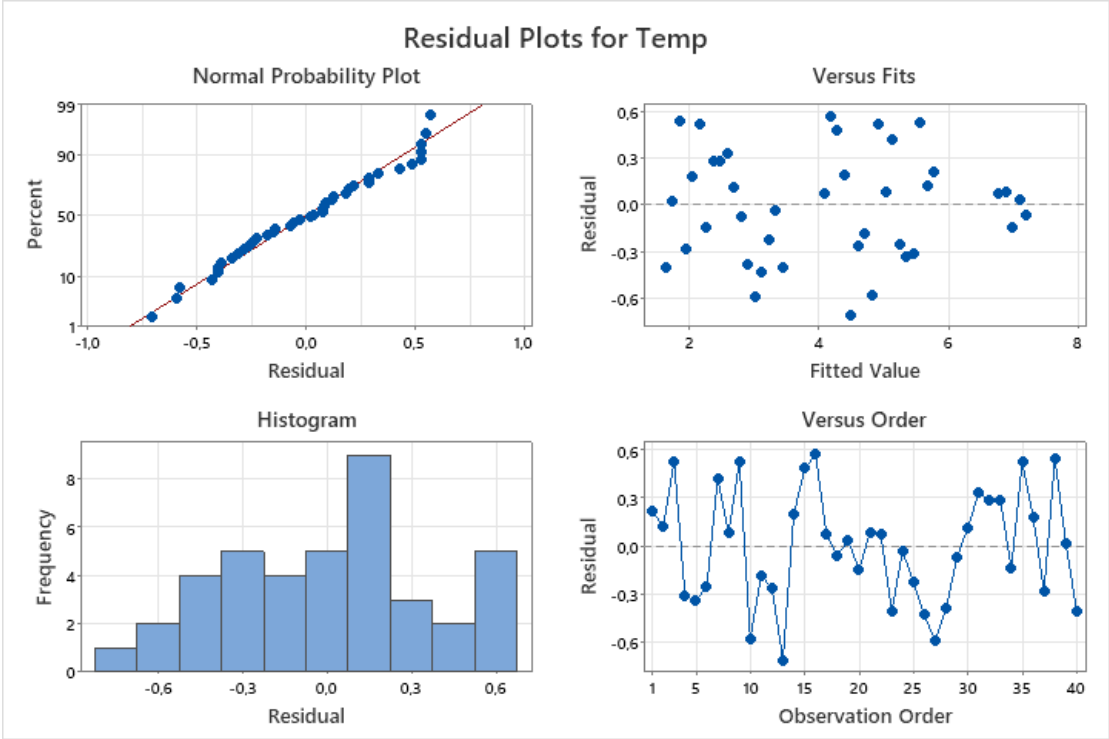
Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0,357465	95,78%	95,55%	95,25%



Point 1 Identify a suitable model.

Assumptions check on residuals:



Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
16	20,95	0,112

The model is adequate.

Exercise 5 (exam 19/07/2019)

A scientific instrument installed on the International Space Station is kept at a low temperature by means of a liquid cooling system. The average temperature measured every day for 40 consecutive days are reported in *temperatures.csv*. From day 18 to day 22 the instrument was used for some extra experiments, whereas normal operating conditions were applied in all the remaining days.

Identify a suitable model.

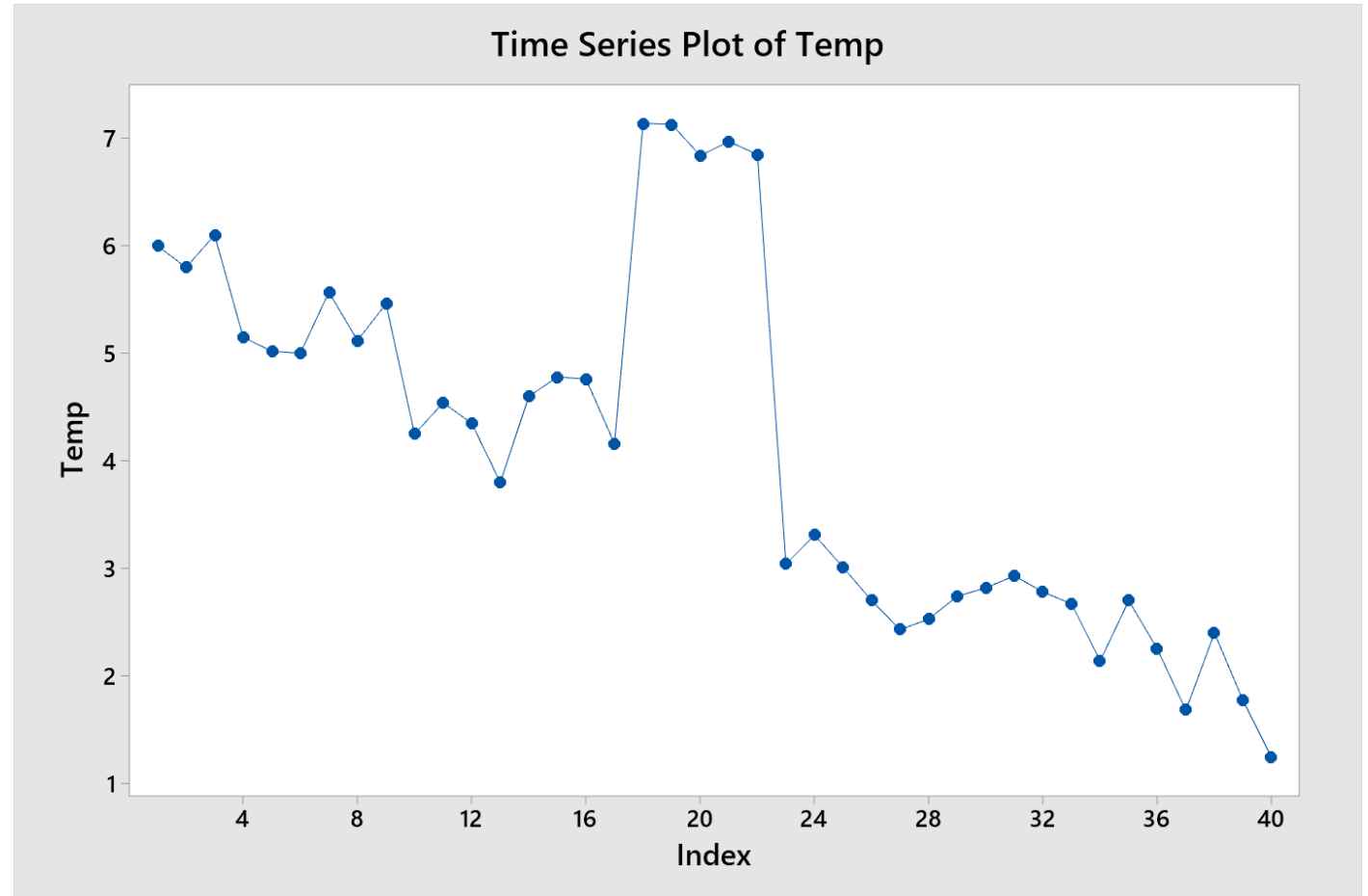
Exercise 5 (solution)

Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

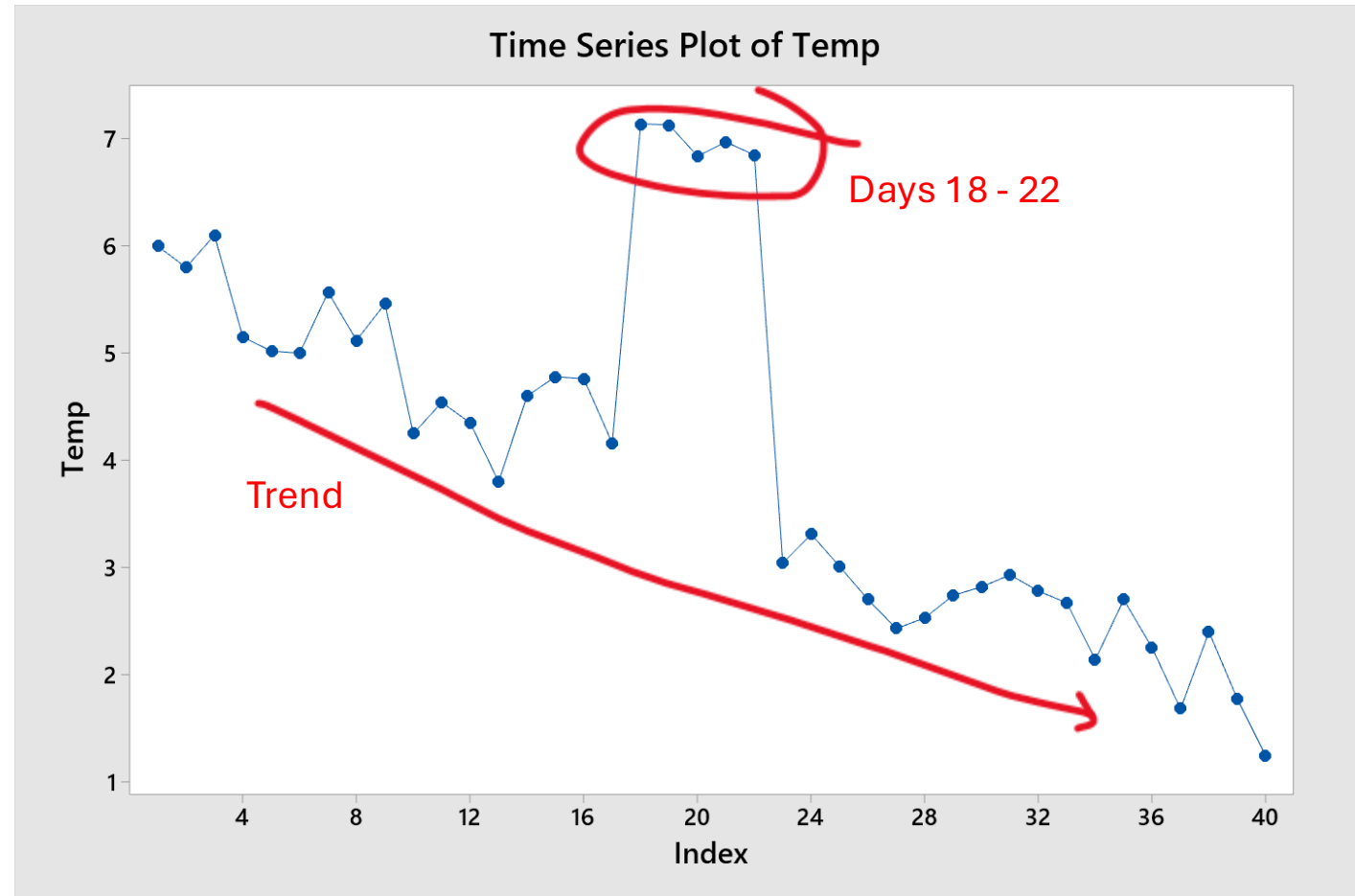
Number of Runs

Observed	Expected	P-Value
4	20,95	0,000

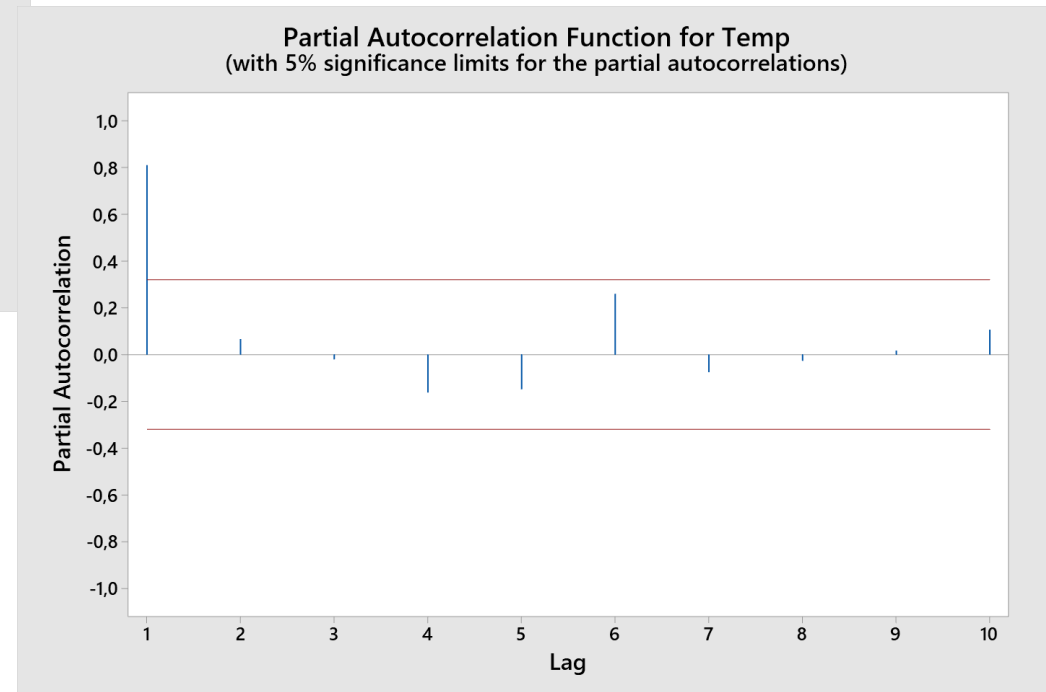
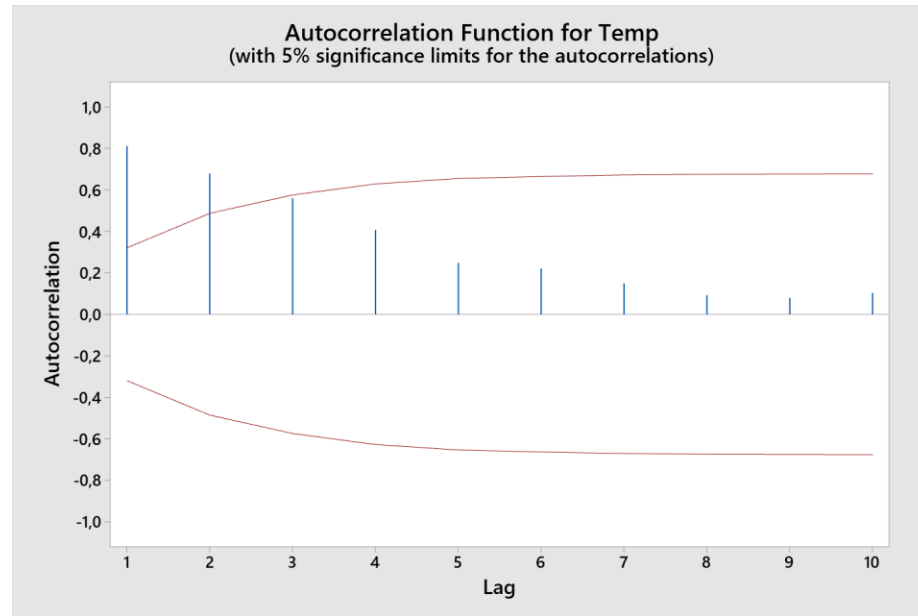


Exercise 5 (solution)

Suitable model: trend + dummy?



Exercise 5 (solution)



Exercise 5 (solution)

$$X_t = \beta_0 + \beta_1 day + \beta_2 dummy + \varepsilon_t$$

Where dummy:

- =1 for days from 18 to 22
- =0 for all other days

C12	C13	C14
Temp	Day	Dummy
6,00	1	0
5,80	2	0
6,10	3	0
5,15	4	0
5,02	5	0
5,00	6	0
5,57	7	0
5,12	8	0
5,46	9	0
4,25	10	0
4,54	11	0
4,35	12	0
3,80	13	0
4,60	14	0
4,78	15	0
4,76	16	0
4,16	17	0
7,14	18	1
7,13	19	1
6,84	20	1
6,97	21	1
6,85	22	1
3,04	23	0
3,31	24	0
3,01	25	0
2,70	26	0
2,43	27	0
2,53	28	0
2,74	29	0
2,82	30	0
2,93	31	0

Exercise 5 (solution)

WORKSHEET 1

Regression Analysis: Temp versus Day; Dummy

Method

Categorical predictor coding (1; 0)

Regression Equation

Dummy

0 Temp = 5,886 - 0,10615 Day

1 Temp = 9,109 - 0,10615 Day

Coefficients

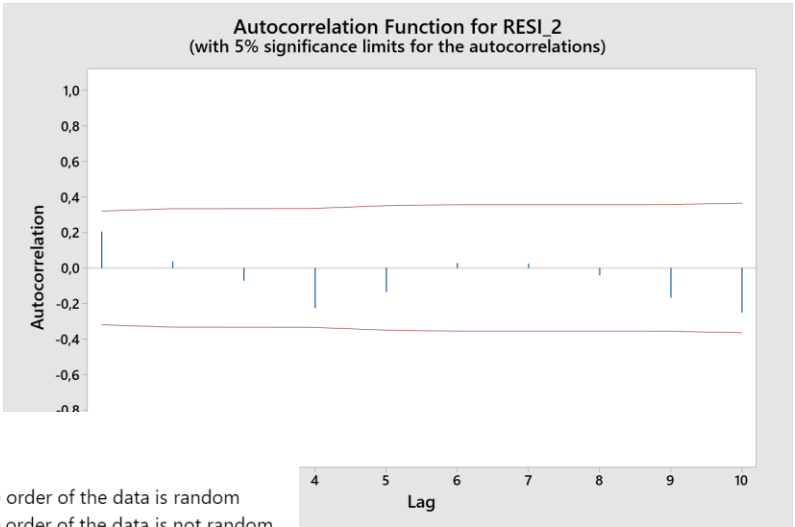
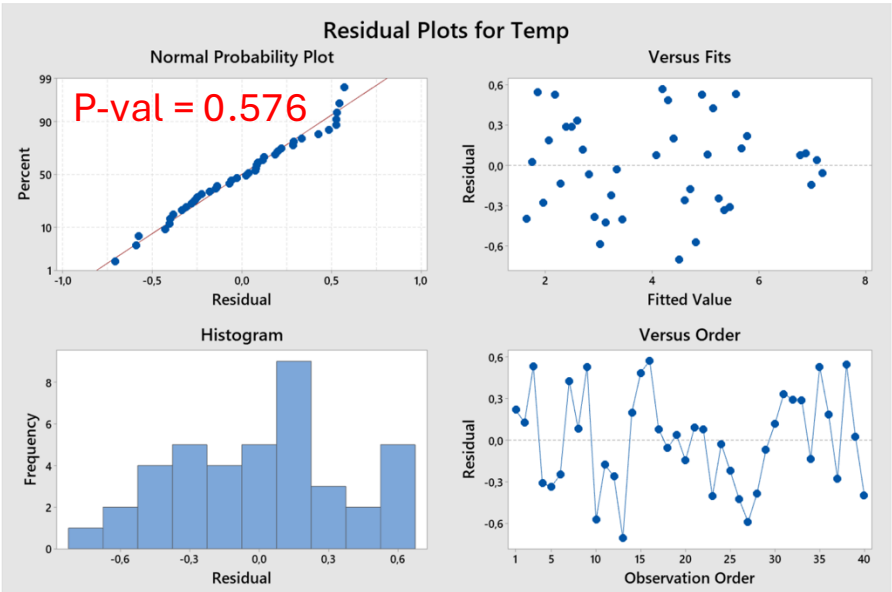
Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	5,886	0,117	50,11	0,000	
Day	-0,10615	0,00490	-21,66	0,000	1,00
Dummy					
1	3,222	0,171	18,85	0,000	1,00

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0,357465	95,78%	95,55%	95,25%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	2	107,197	53,5987	419,46	0,000
Day	1	60,039	60,0391	469,86	0,000
Dummy	1	45,420	45,4197	355,45	0,000
Error	37	4,728	0,1278		
Total	39	111,925			



Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
16	20,95	0,112

Exercise 5 (solution)

